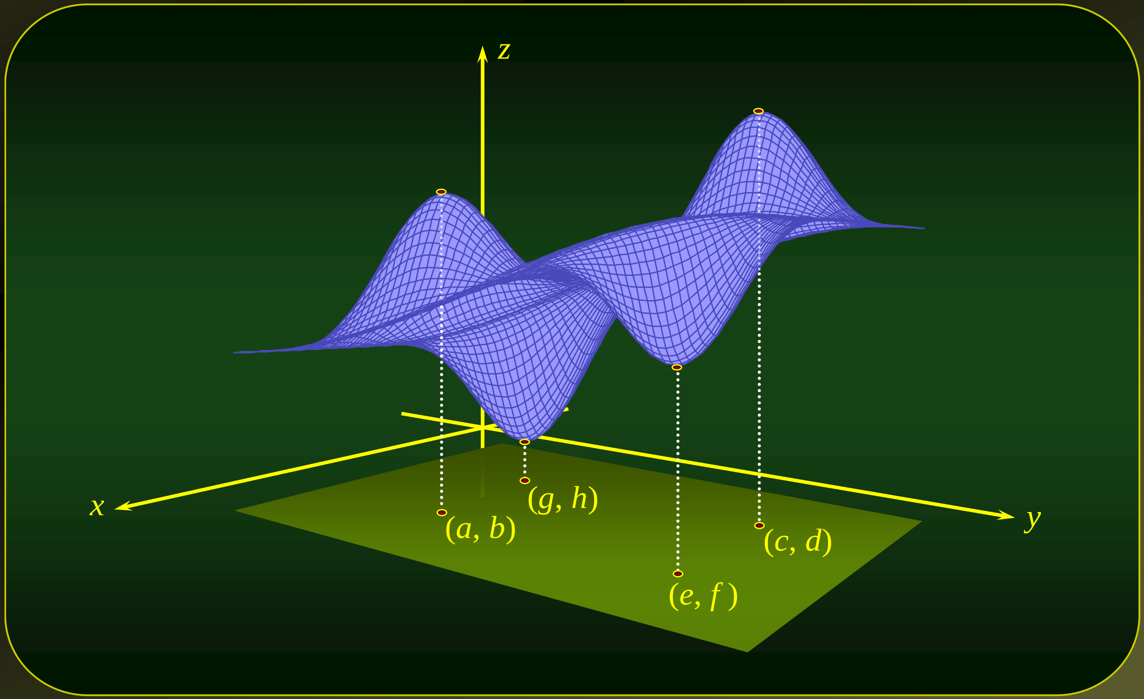
Maxima and Minima
of Functions of Several Variables

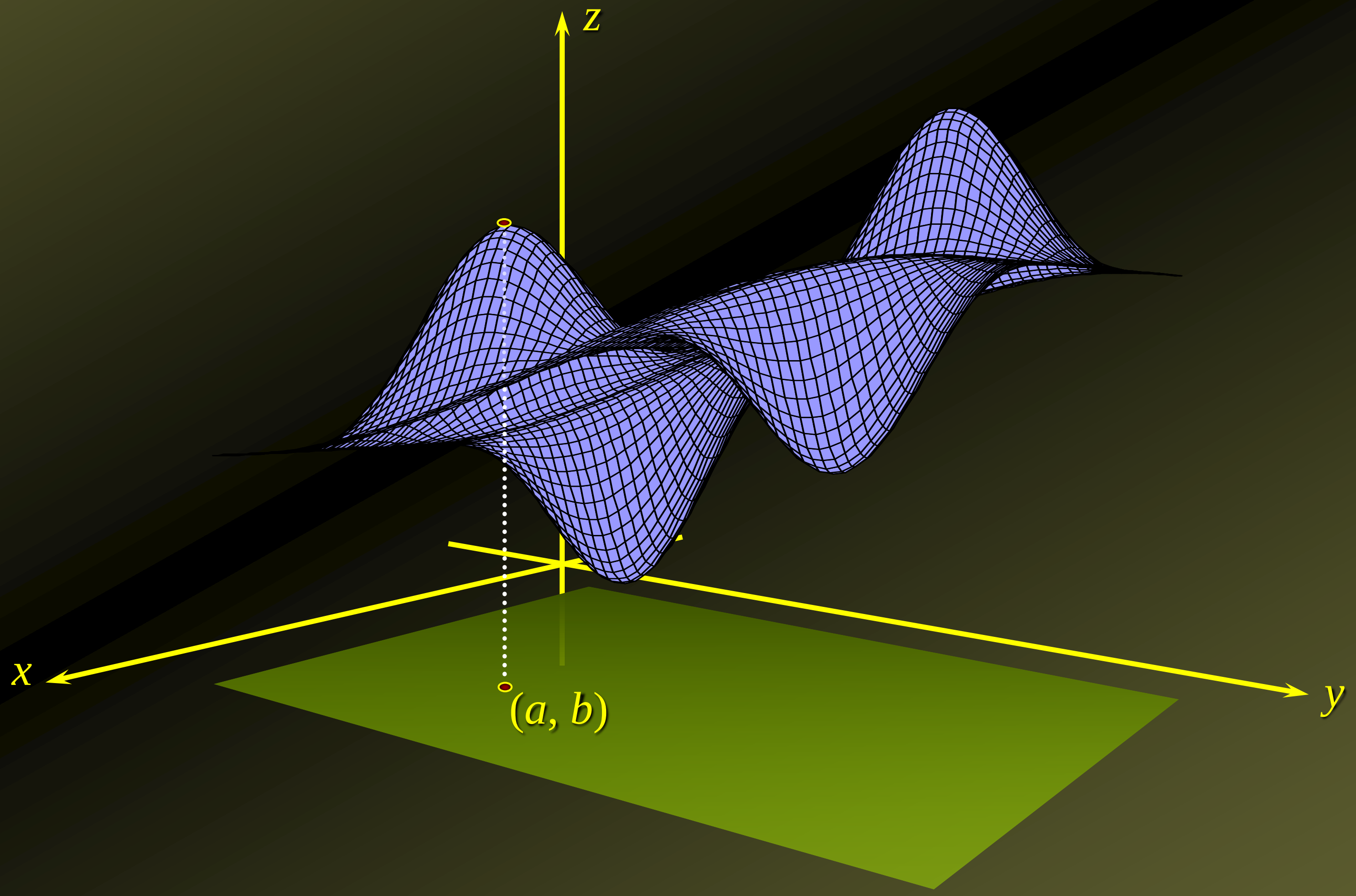


Relative Extrema of a Function of Two Variables

- Let f be a function defined on a region R containing the point (a, b).
- Then, f has a relative maximum at (a, b) if $f(x, y) \leq f(a, b)$ for all points (x, y) that are sufficiently close to (a, b).
  - The number $f(a, b)$ is called a relative maximum value.
- Similarly, f has a relative minimum at (a, b) if $f(x, y) \geq f(a, b)$ for all points (x, y) that are sufficiently close to (a, b).
  - The number $f(a, b)$ is called a relative minimum value.

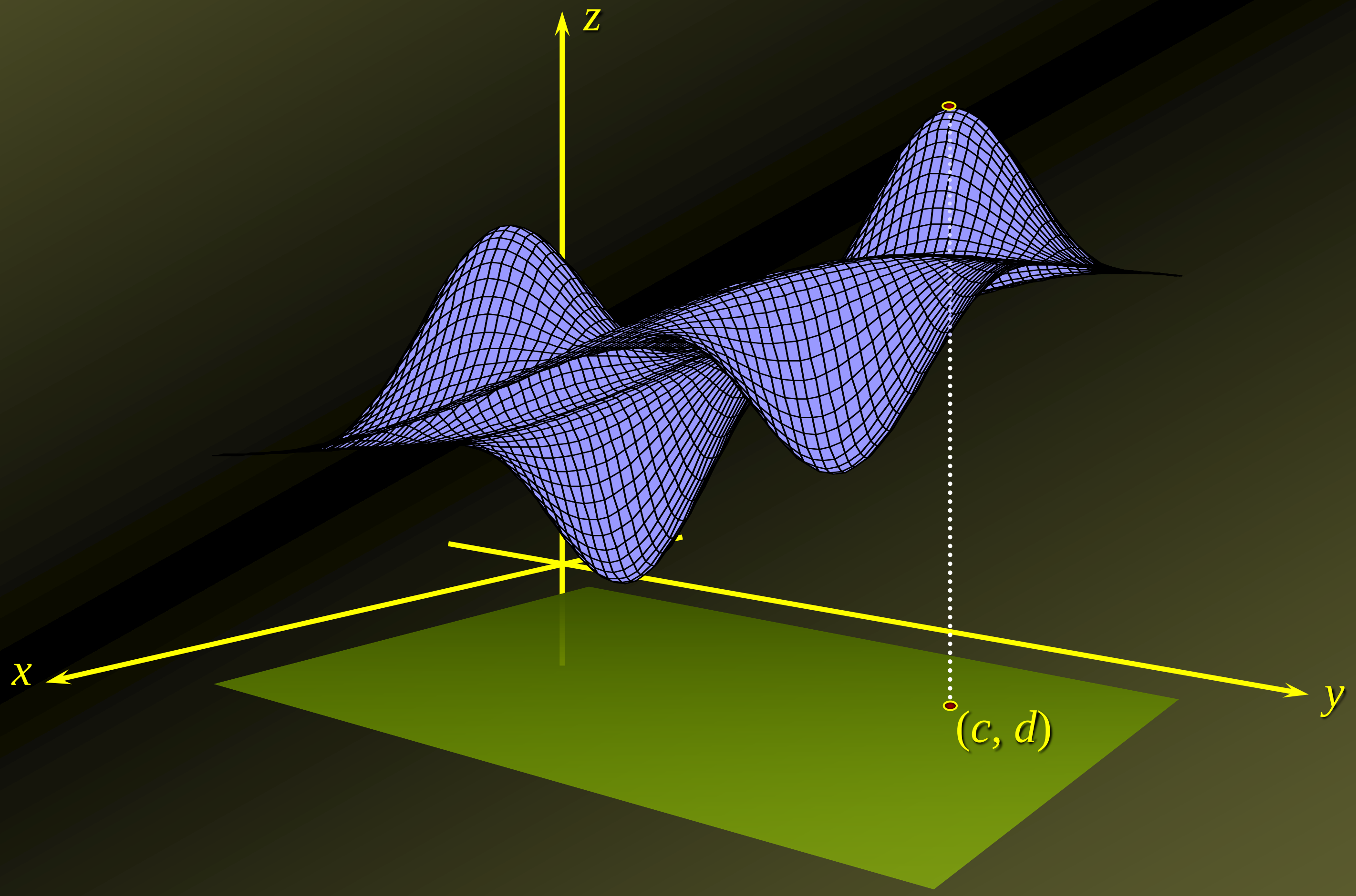
Graphic Example

- **There is a relative maximum at** (a, b)**.**



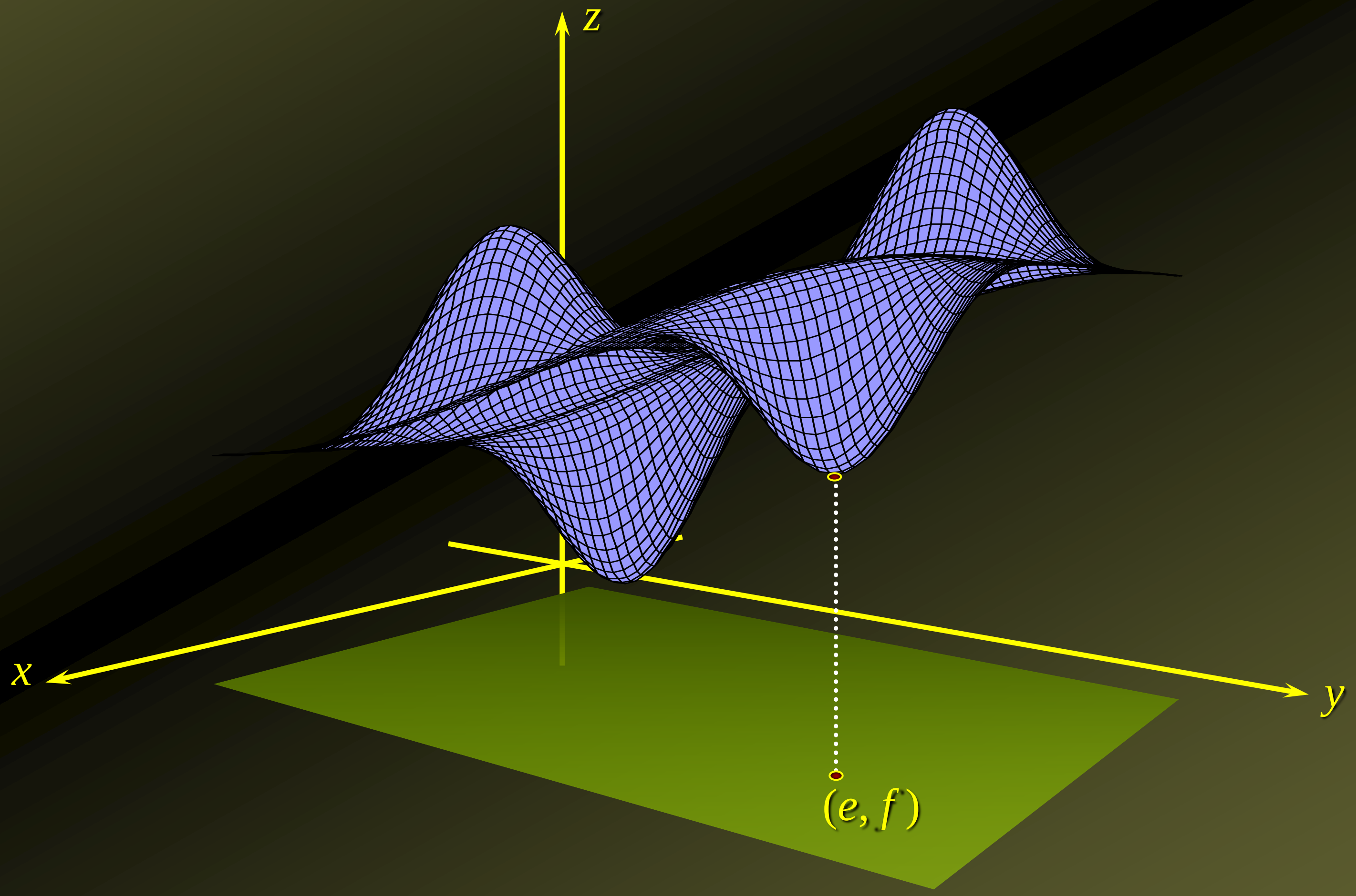
Graphic Example

- There is an **absolute maximum** at (c, d).
  (It is also a **relative maximum**)



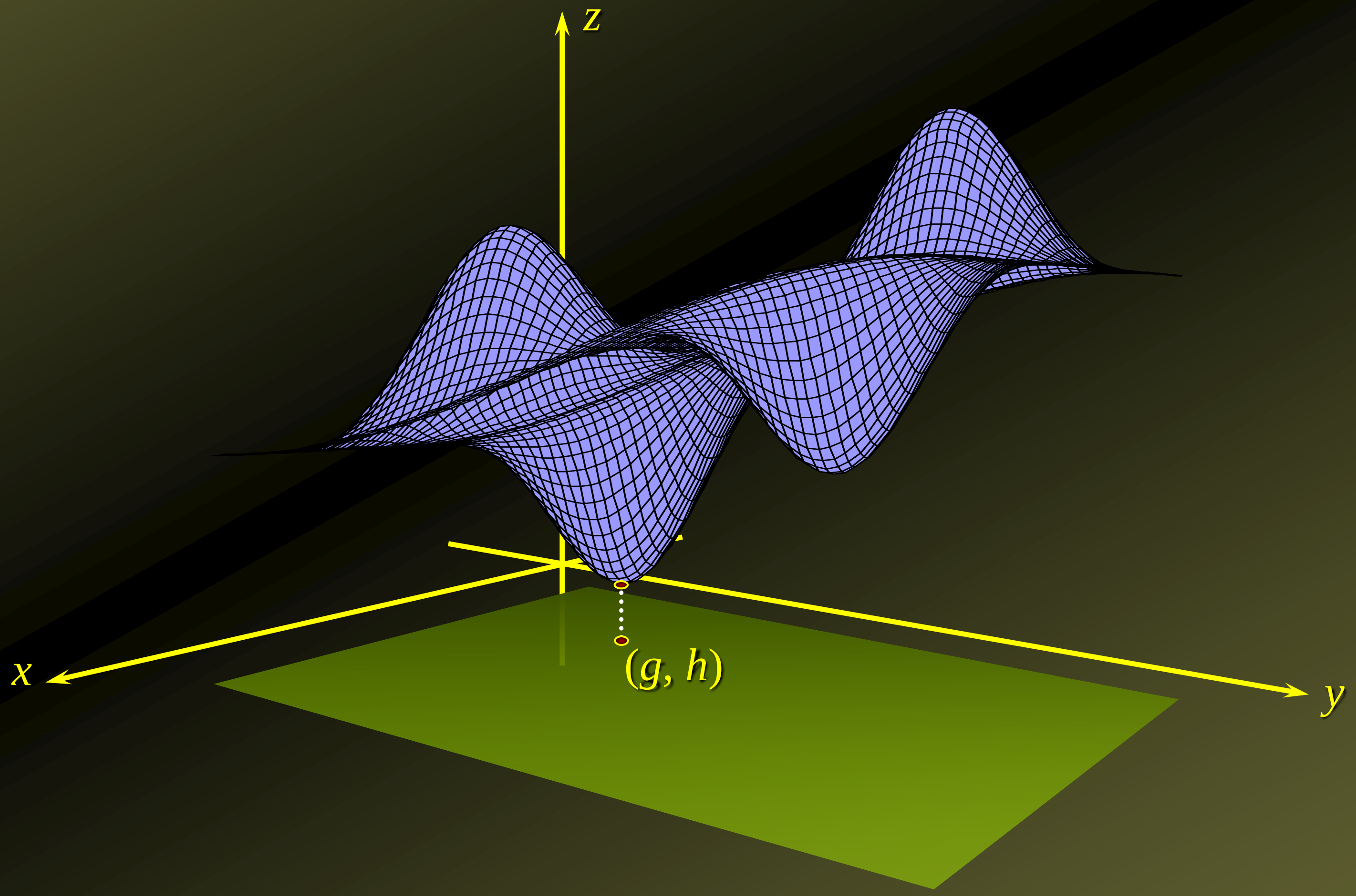
Graphic Example

- There is a **relative minimum** at (e, f).



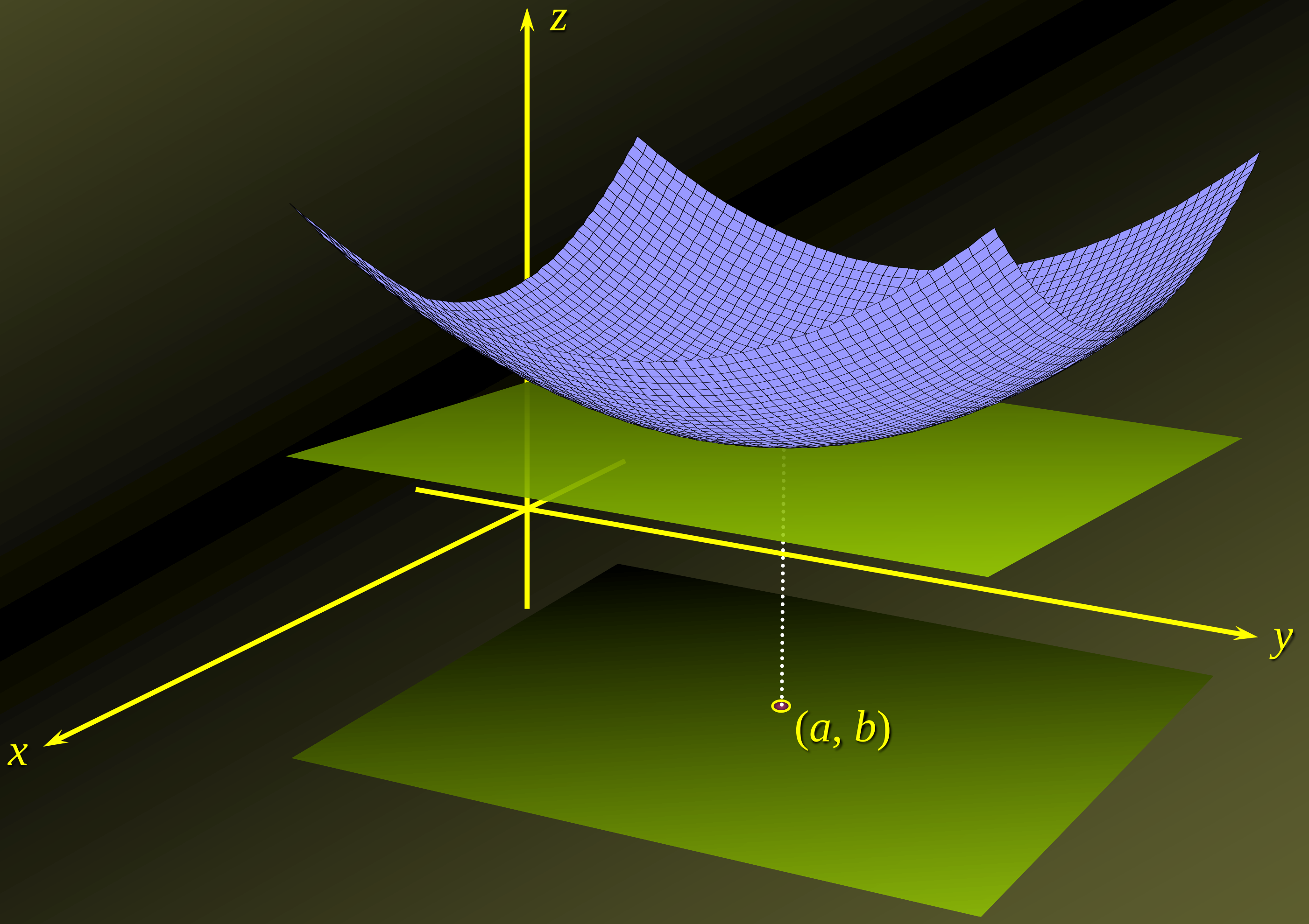
Graphic Example

- There is an **absolute minimum** at (g, h).
  (It is also a **relative minimum**)



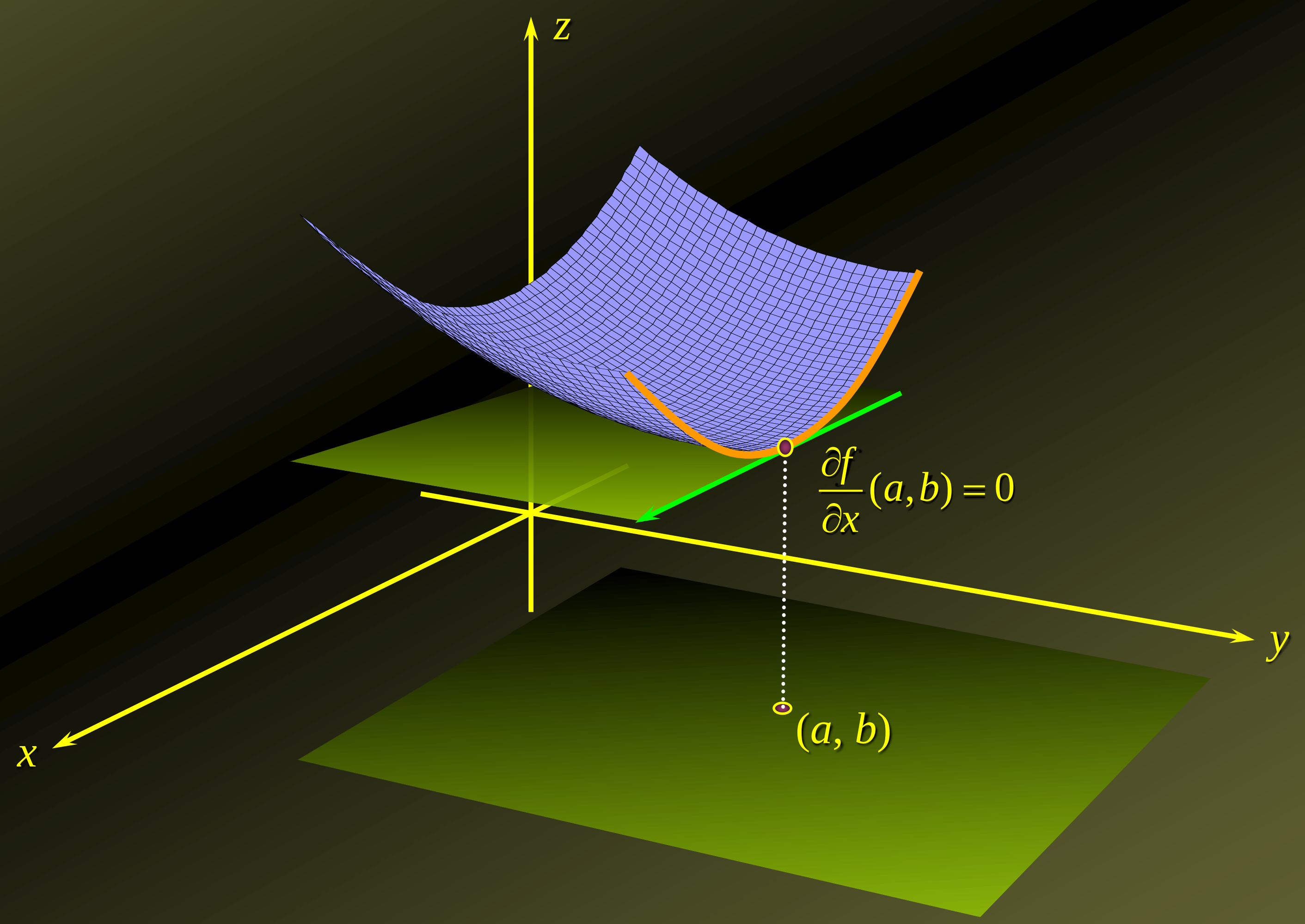
Relative Minima

- At a minimum point of the graph of a function of two variables, such as point (a, b) below, the plane tangent to the graph of the function is horizontal (assuming the surface of the graph is smooth):



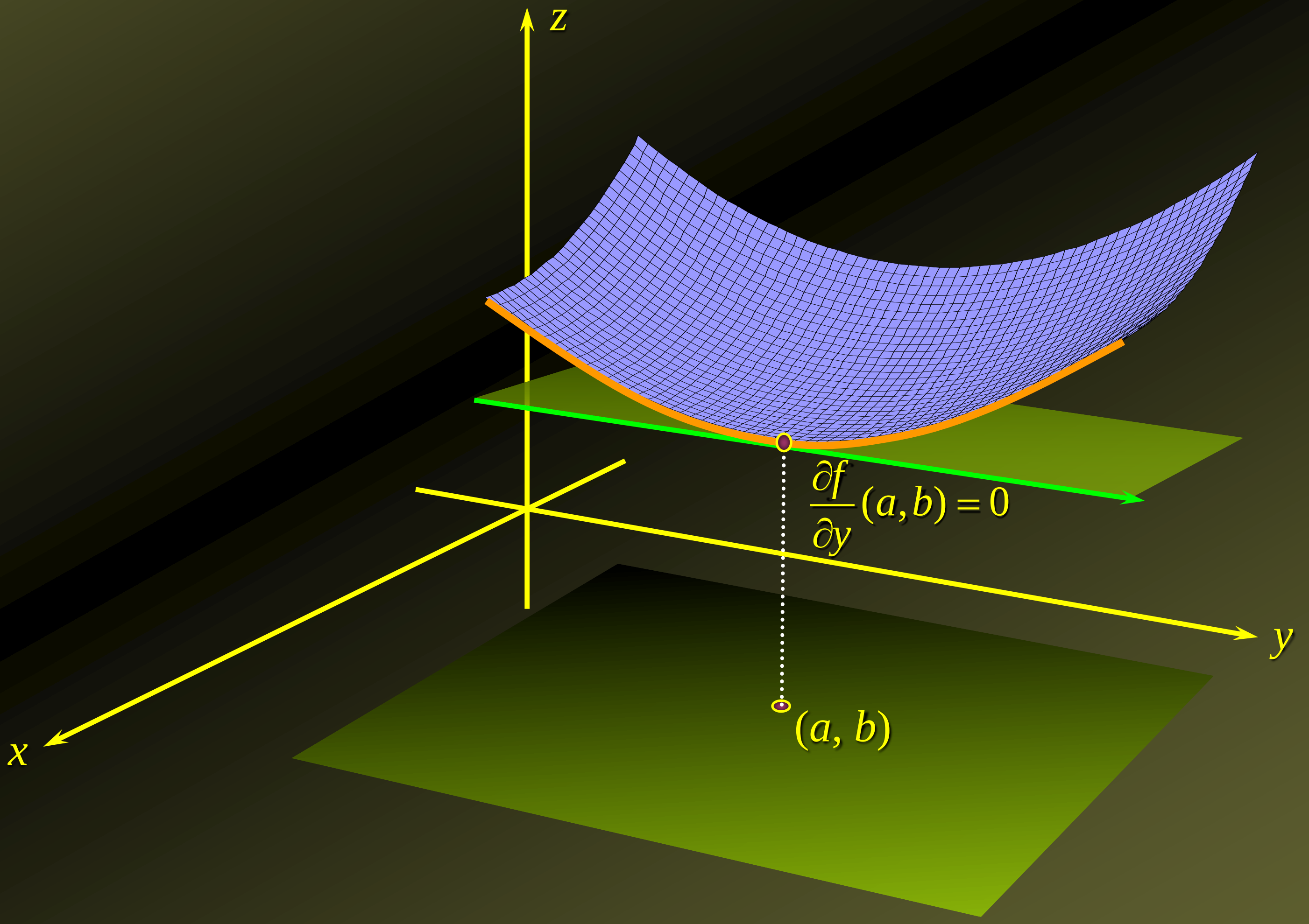
Relative Minima

- Thus, at a **minimum point**, the graph of the function has a **slope of zero** along a direction **parallel** to the ***x*-axis**:



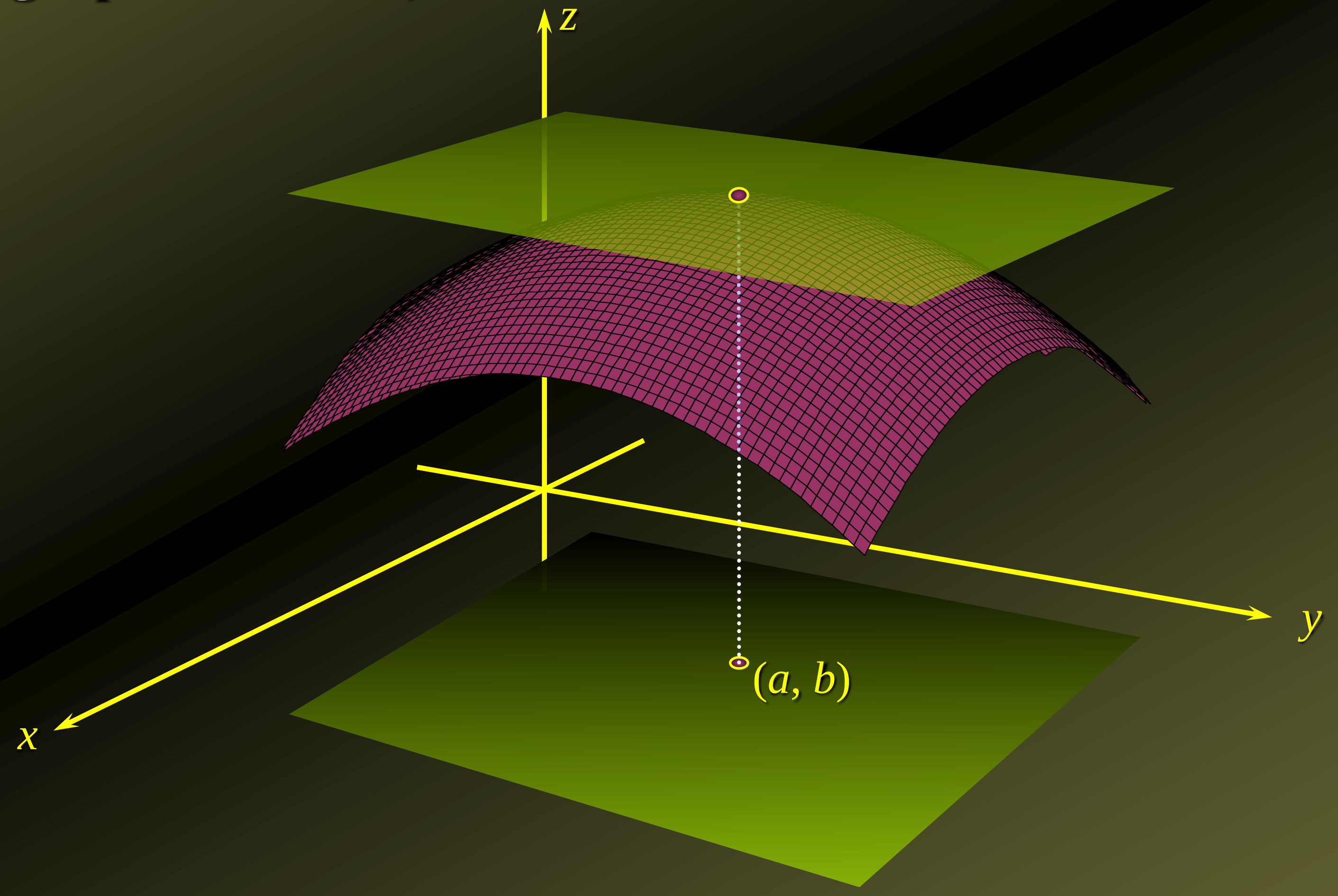
Relative Minima

- Similarly, at a **minimum point**, the graph of the function has a **slope of zero** along a direction **parallel** to the ***y*-axis**:



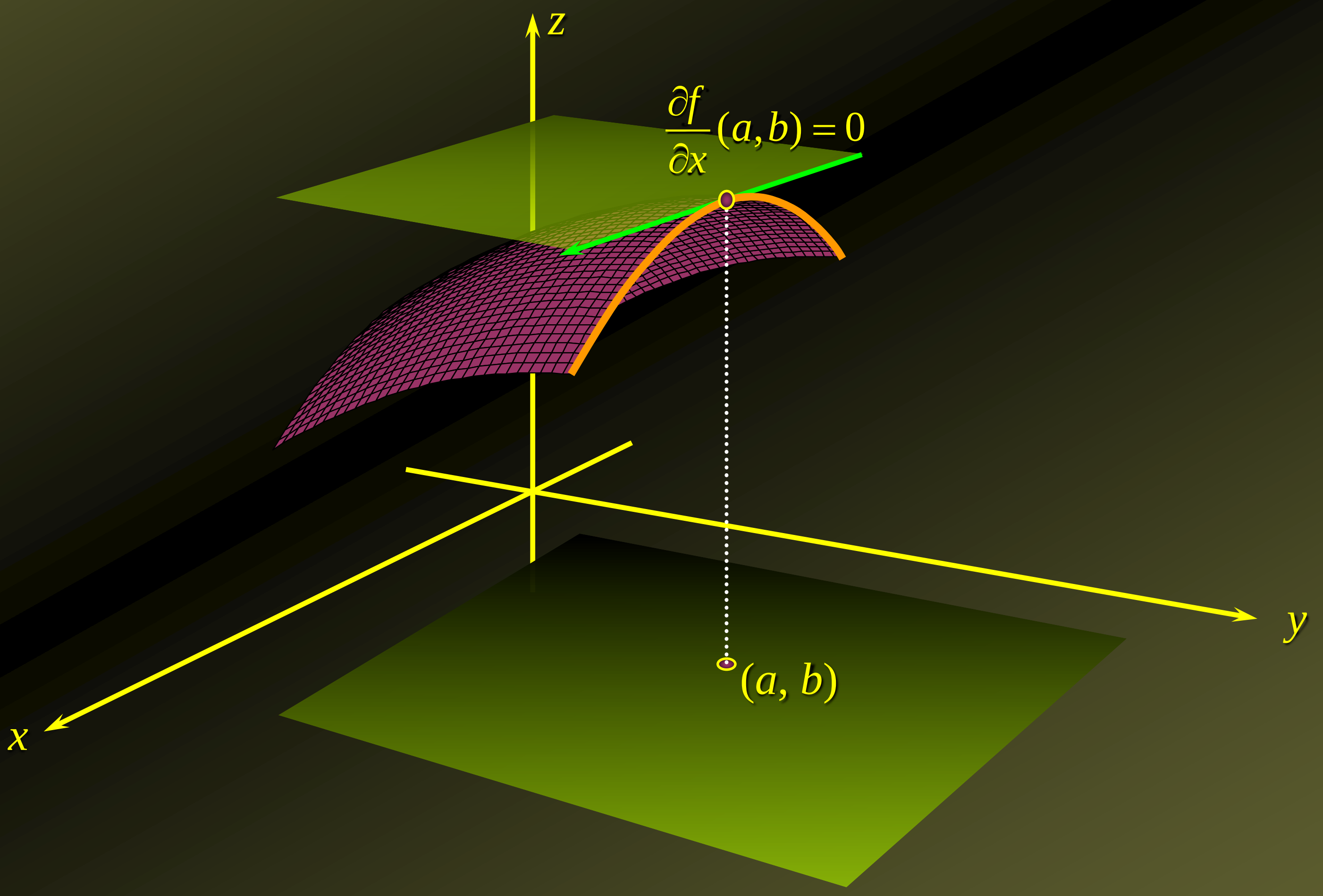
Relative Maxima

- At a maximum point of the graph of a function of two variables, such as point (a, b) below, the plane tangent to the graph of the function is horizontal (assuming the surface of the graph is smooth):



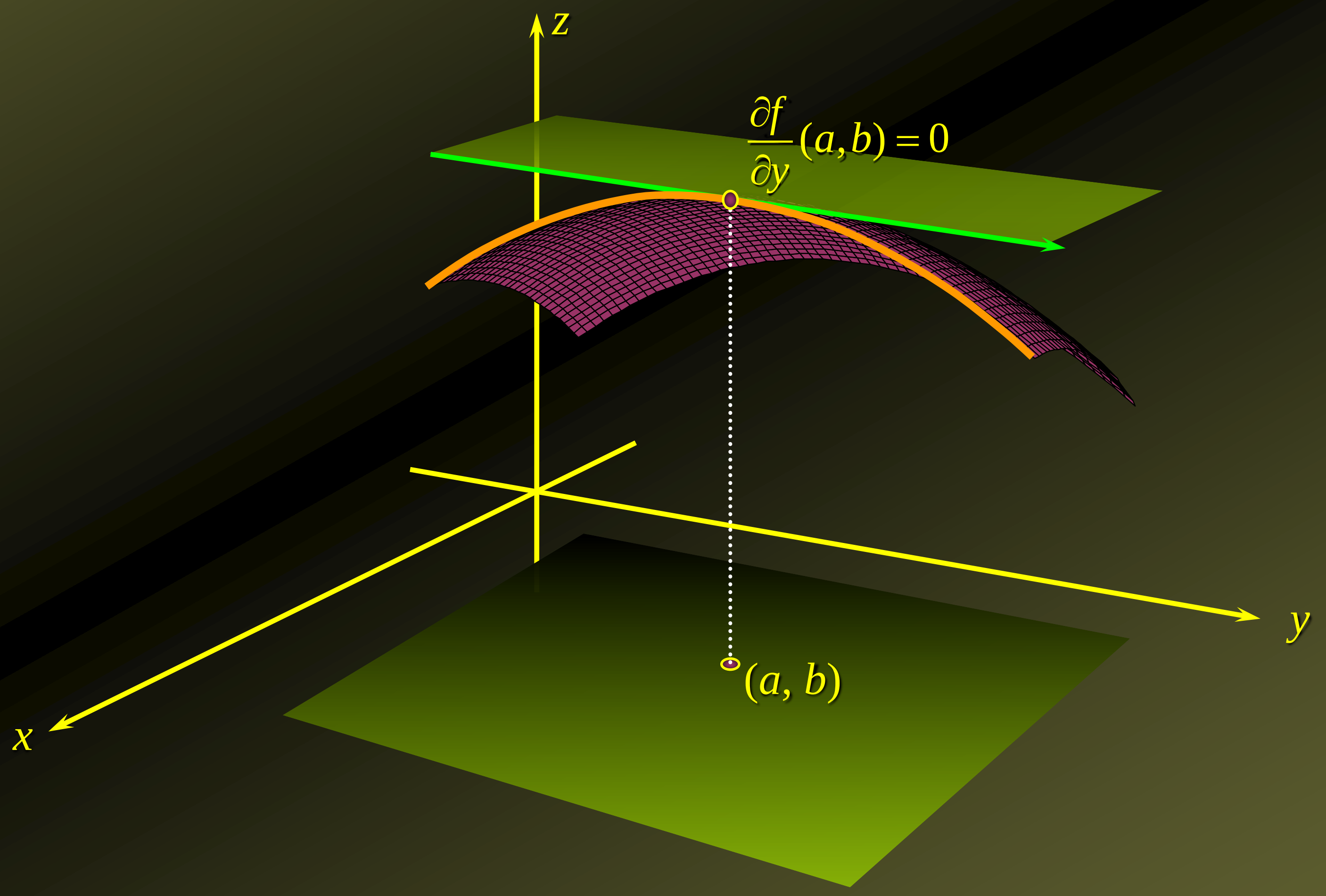
Relative Maxima

- Thus, at a **maximum point**, the graph of the function has a **slope of zero** along a direction **parallel** to the *x*-axis:



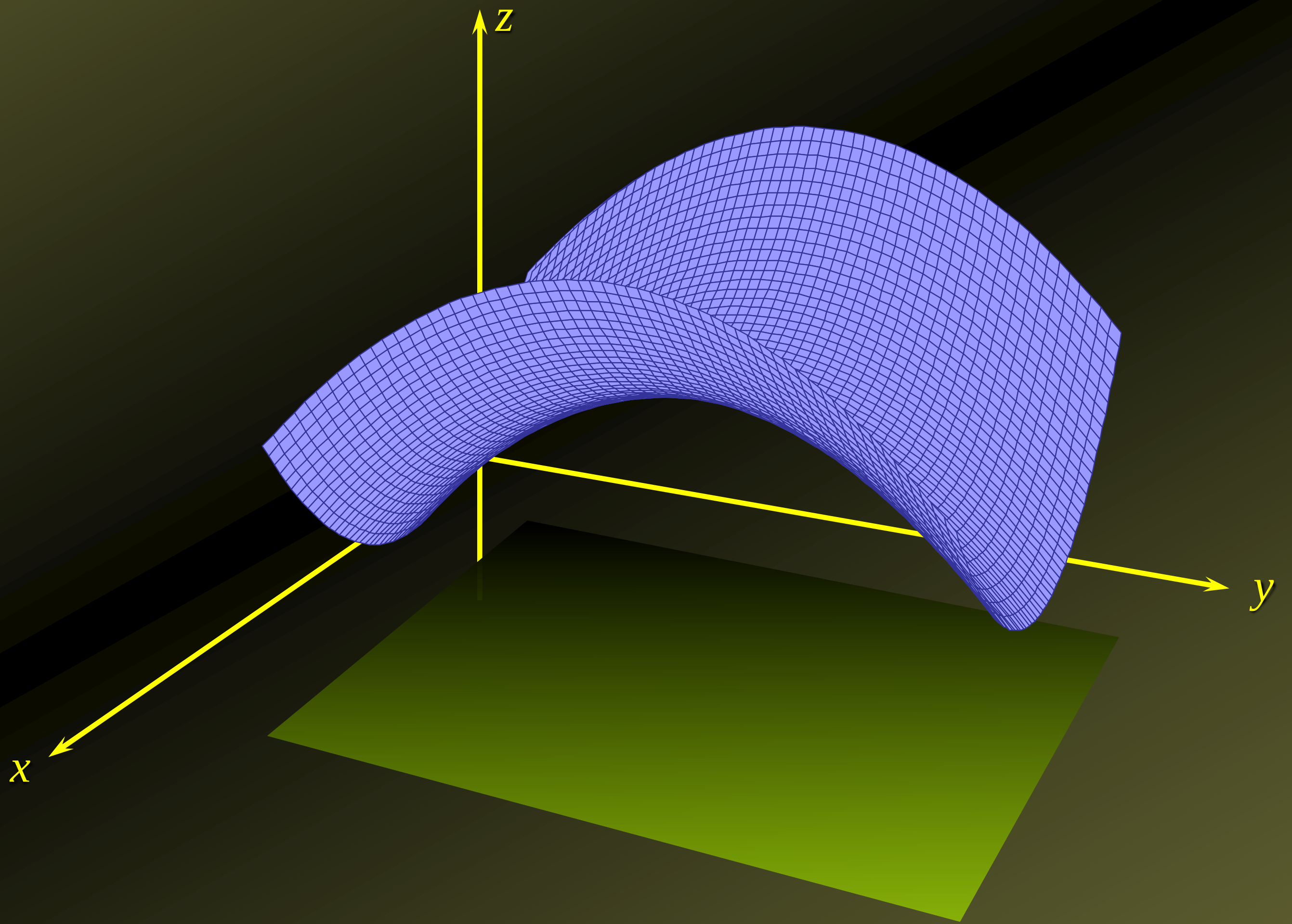
Relative Maxima

- Similarly, at a **maximum point**, the graph of the function has a **slope of zero** along a direction **parallel** to the y**-axis**:



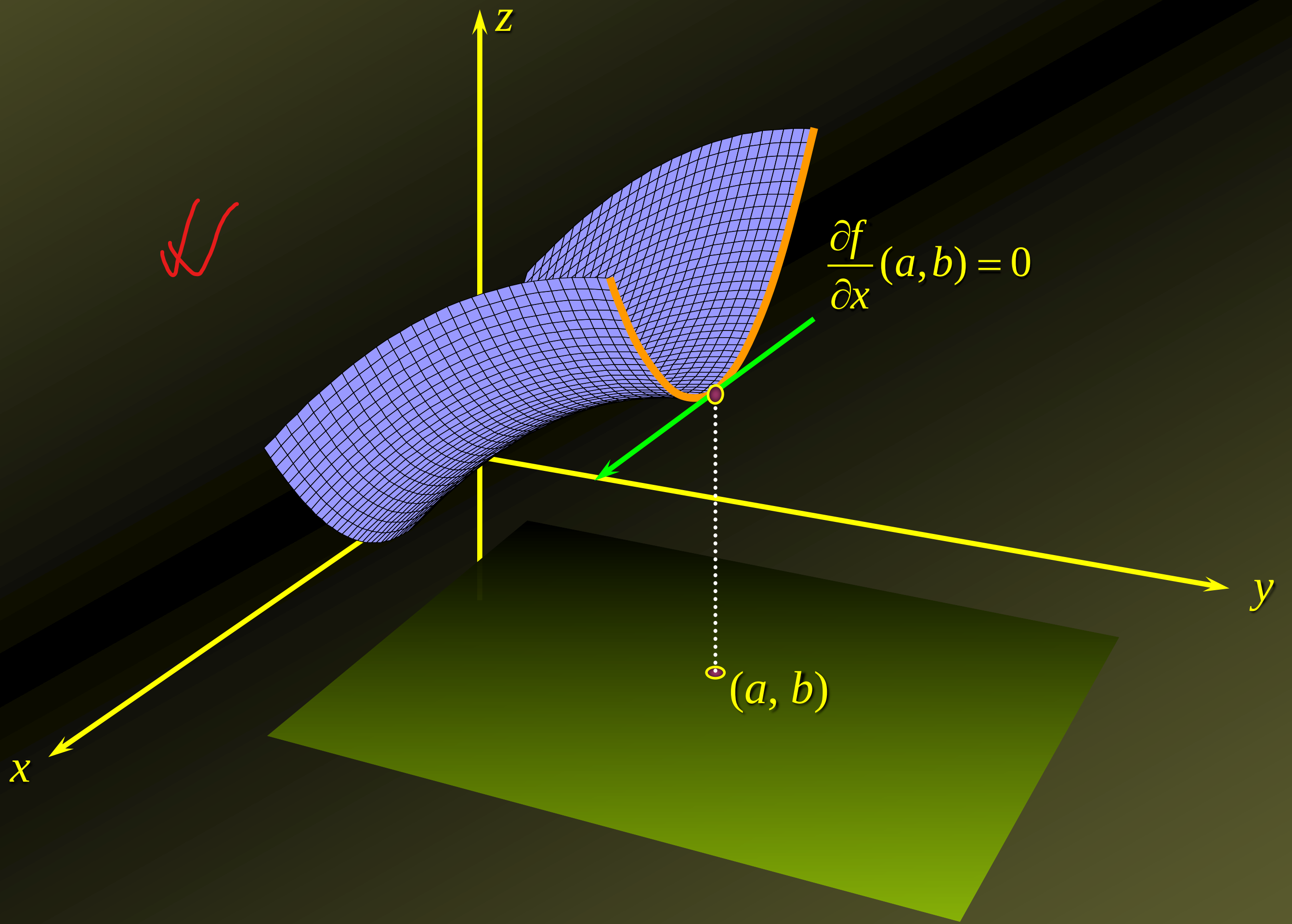
Saddle Point

- In the case of a saddle point, both partials are equal to zero, but the point is neither a maximum nor a minimum.



Saddle Point

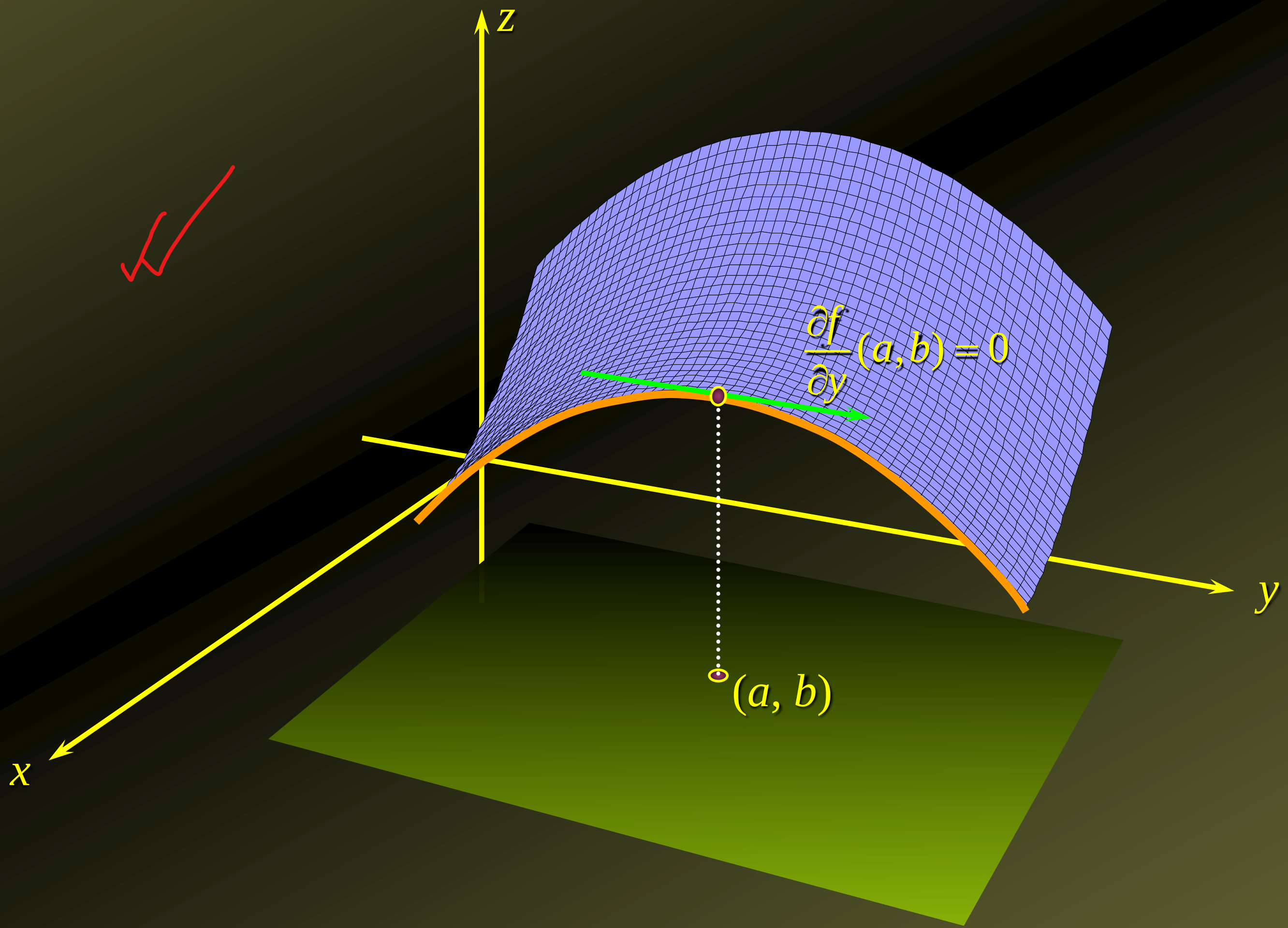
In the case of a saddle point, the function is at a minimum along one vertical plane…



Saddle Point

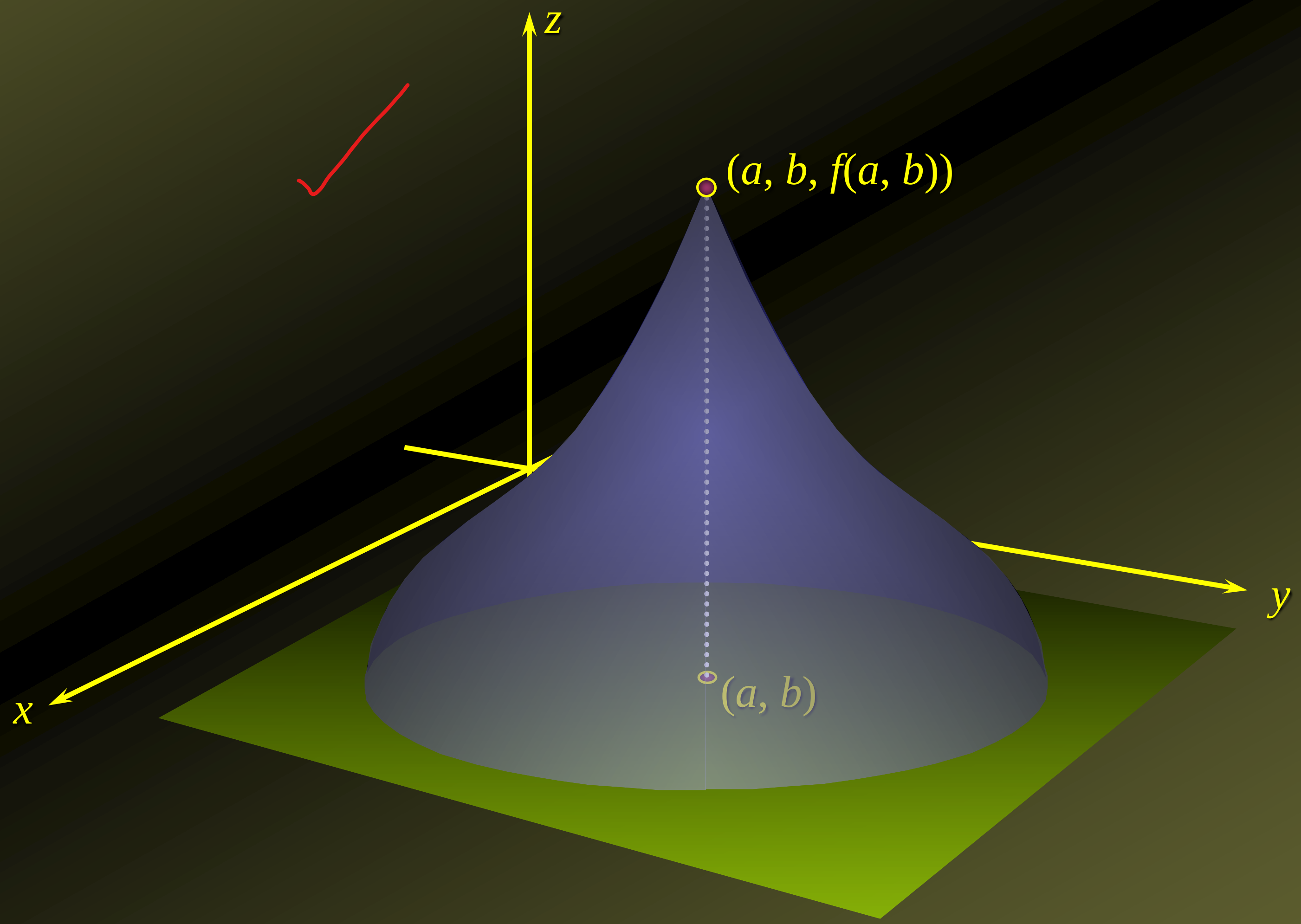
… but at a maximum along the ~~perpendicular vertical~~ plane.

Horizontal



Extrema When Partial Derivatives are Not Defined

- A maximum (or minimum) may also occur when both partial derivatives are not defined, such as point (a, b) in the graph below:



Critical Point of a Function

- A critical point of f is a point (a, b) in the domain of f such that both

$$\boxed{\frac{\partial f}{\partial x}(a,b) = 0} \quad \text{and} \quad \boxed{\frac{\partial f}{\partial y}(a,b) = 0}$$

or at least one of the partial derivatives does not exist.

Determining Relative Extrema

1. **Find the critical points of $f(x, y)$ by solving the system of simultaneous equations**

$$f_x = 0 \qquad\qquad f_y = 0$$

2. **The second derivative test: Let**

$$D(x, y) = f_{xx} f_{yy} - f^2_{xy}$$

3. **Then,**
   a. $D(a, b) > 0$ **and** $f_{xx}(a, b) < 0$ **implies that** $f(x, y)$ **has a relative maximum at the point** (a, b)**.**
   b. $D(a, b) > 0$ **and** $f_{xx}(a, b) > 0$ **implies that** $f(x, y)$ **has a relative minimum at the point** (a, b)**.**
   c. $D(a, b) < 0$ **implies that** $f(x, y)$ **has neither a relative maximum nor a relative minimum at the point** (a, b)**, it has instead a saddle point.**
   d. $D(a, b) = 0$ **implies that the test is inconclusive, so some other technique must be used to solve the problem.**

Examples

- Find the relative extrema of the function

$$f(x, y) = x^2 + y^2$$

Solution

- We have $f_x = 2x$ and $f_y = 2y$.
- To find the critical points, we set $f_x = 0$ and $f_y = 0$ and solve the resulting system of simultaneous equations

$$2x = 0 \quad \text{and} \quad 2y = 0$$

obtaining $x = 0$, $y = 0$, or $(0, 0)$, as the sole critical point.

- Next, apply the second derivative test to determine the nature of the critical point $(0, 0)$.
- We compute $f_{xx} = 2$, $f_{yy} = 2$, and $f_{xy} = 0$,
- Thus, $D(x, y) = f_{xx} f_{yy} - f^2_{xy} = (2)(2) - (0)^2 = 4.$

Examples

- Find the relative extrema of the function

$$f(x, y) = x^2 + y^2$$

Solution

- We have $D(x, y) = 4$, and in particular, $D(0, 0) = 4$.
- Since $D(0, 0) > 0$ and $f_{xx} = 2 > 0$, we conclude that f has a relative minimum at the point $(0, 0)$.
- The relative minimum value, $f(0, 0) = 0$, also happens to be the absolute minimum of f.

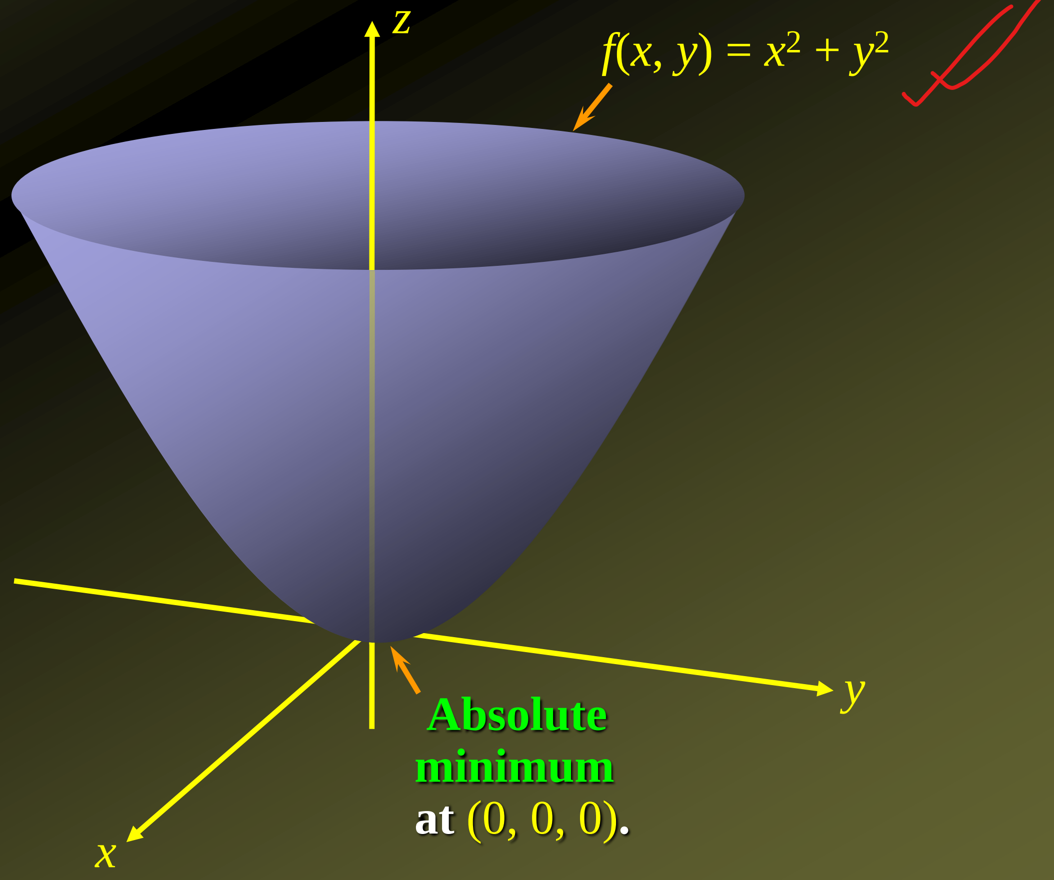
Examples

- Find the relative extrema of the function

$$f(x,y) = x^2 + y^2$$

Solution

- The relative minimum value, $f(0, 0) = 0$, also happens to be the absolute minimum of f:



Examples

- Find the relative extrema of the function

$$f(x, y) = 3x^2 - 4xy + 4y^2 - 4x + 8y + 4$$

Solution

- We have $f_x = 6x - 4y - 4$ and $f_y = -4x + 8y + 8$
- To find the critical points, we set $f_x = 0$ and $f_y = 0$ and solve the resulting system of simultaneous equations

$$6x - 4y - 4 = 0 \quad \text{and} \quad -4x + 8y + 8 = 0$$

obtaining $x = 0$, $y = -1$, or $(0, -1)$, as the sole critical point.

- Next, apply the second derivative test to determine the nature of the critical point $(0, -1)$.
- We compute $f_{xx} = 6$, $f_{yy} = 8$, and $f_{xy} = -4$,
- Thus, $D(x, y) = f_{xx} \cdot f_{yy} - f^2_{xy} = (6)(8) - (-4)^2 = 32.$

Examples

- Find the relative extrema of the function

$$f(x, y) = 3x^2 - 4xy + 4y^2 - 4x + 8y + 4$$

Solution

- We have $D(x, y) = 32$, and in particular, $D(0, -1) = 32$.
- Since $D(0, -1) > 0$ and $f_{xx} = 6 > 0$, we conclude that f has a relative minimum at the point $(0, -1)$.
- The relative minimum value, $f(0, -1) = 0$, also happens to be the absolute minimum of f.

Examples

- Find the relative extrema of the function

$$f(x,y) = 4y^3 + x^2 - 12y^2 - 36y + 2$$

Solution

- We have $f_x = 2x$ and $f_y = 12y^2 - 24y - 36$
- To find the critical points, we set $f_x = 0$ and $f_y = 0$ and solve the resulting system of simultaneous equations

$$2x = 0 \quad \text{and} \quad 12y^2 - 24y - 36 = 0$$

- The first equation implies that $x = 0$, while the second equation implies that $y = -1$ or $y = 3$.
- Thus, there are two critical points of f: $(0, -1)$ and $(0, 3)$.
- To apply the second derivative test, we calculate

$$f_{xx} = 2 \qquad f_{yy} = 24(y-1) \qquad f_{xy} = 0$$

$$D(x, y) = \boxed{f_{xx} \cdot f_{yy} - f^2_{xy}} = (2) \cdot 24(y-1) - (0)^2 = 48(y-1)$$

Examples

- Find the relative extrema of the function

$$f(x,y) = 4y^3 + x^2 - 12y^2 - 36y + 2$$

Solution

- Apply the second derivative test to the critical point $(0, -1)$:
- We have $D(x, y) = 48(y - 1)$.
- In particular, $D(0, -1) = 48[(-1) - 1] = -96$.
- Since $D(0, -1) = -96 < 0$ we conclude that f has a saddle point at $(0, -1)$.
- The saddle point value is $f(0, -1) = 22$, so there is a saddle point at $(0, -1, 22)$.

Examples

- Find the relative extrema of the function

$$f(x, y) = 4y^3 + x^2 - 12y^2 - 36y + 2$$

Solution

- Apply the second derivative test to the critical point (0, 3):
- We have $D(x, y) = 48(y - 1)$.
- In particular, $D(0, 3) = 48[(3) - 1] = 96$.
- Since $D(0, -1) = 96 > 0$ and $f_{xx}(0, 3) = 2 > 0$, we conclude that f has a relative minimum at the point (0, 3).
- The relative minimum value, $f(0, 3) = -106$, so there is a relative minimum at (0, 3, –106).

Applied Example: Maximizing Profit

- The total weekly revenue that Acrosonic realizes in producing and selling its loudspeaker system is given by

$$R(x,y) = -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 300x + 240y$$

where x denotes the number of fully assembled units and y denotes the number of kits produced and sold each week.

- The total weekly cost attributable to the production of these loudspeakers is

$$C(x,y) = 180x + 140y + 5000$$

- Determine how many assembled units and how many kits should be produced per week to maximize profits.

Applied Example: Maximizing Profit

Solution

- The contribution to Acrosonic's weekly profit stemming from the production and sale of the bookshelf loudspeaker system is given by

$$
\begin{aligned}
P(x, y) &= R(x, y) - C(x, y) \\
&= \left(-\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 300x + 240y\right) - (180x + 140y + 5000) \\
&= -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 120x + 100y - 5000
\end{aligned}
$$

Applied Example: Maximizing Profit

Solution

- We have $P(x, y) = -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 120x + 100y - 5000$
- To find the relative maximum of the profit function P, we first locate the critical points of P.
- Setting P_x and P_y equal to zero, we obtain

$$P_x = -\frac{1}{2}x - \frac{1}{4}y + 120 = 0 \quad \text{and} \quad P_y = -\frac{3}{4}y - \frac{1}{4}x + 100 = 0$$

- Solving the system of equations we get $x = 208$ and $y = 64$.
- Therefore, P has only one critical point at $(208, 64)$.

Applied Example: Maximizing Profit

Solution

- To test if the point (208, 64) is a solution to the problem, we use the second derivative test.
- We compute

$$P_{xx} = -\frac{1}{2} \qquad P_{yy} = -\frac{3}{4} \qquad P_{xy} = -\frac{1}{4}$$

- So,

$$D(x, y) = \left(-\frac{1}{2}\right)\left(-\frac{3}{4}\right) - \left(-\frac{1}{4}\right)^2 = \frac{3}{8} - \frac{1}{16} = \frac{5}{16}$$

- In particular, $D(208, 64) = 5/16 > 0$.
- Since $D(208, 64) > 0$ and $P_{xx}(208, 64) < 0$, the point (208, 64) yields a relative maximum of P.

Applied Example: Maximizing Profit

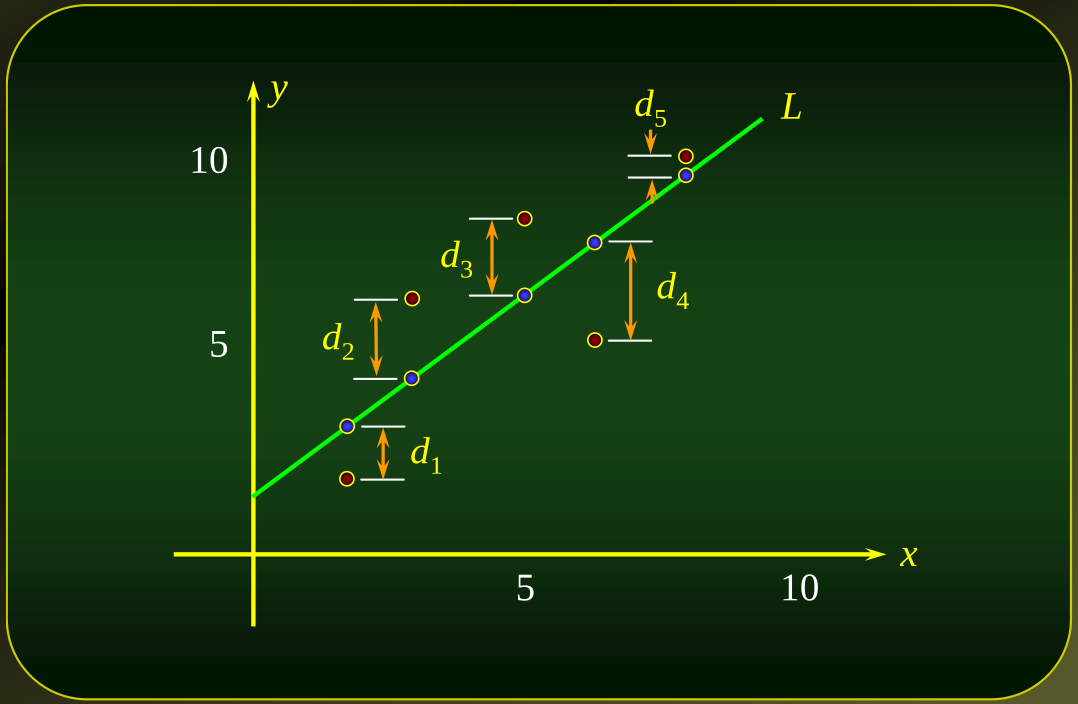
Solution

- The relative maximum at (208, 64) is also the absolute maximum of P.
- We conclude that Acrosonic can maximize its weekly profit by manufacturing 208 assembled units and 64 kits.
- The maximum weekly profit realizable with this output is

$$P(x, y) = -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 120x + 100y - 5000$$

$$\begin{aligned} P(208, 64) &= -\frac{1}{4}(208)^2 - \frac{3}{8}(64)^2 - \frac{1}{4}(208)(64) \\ &\quad + 120(208) + 100(64) - 5000 \\ &= \$10,680 \end{aligned}$$

The Method of Least Squares

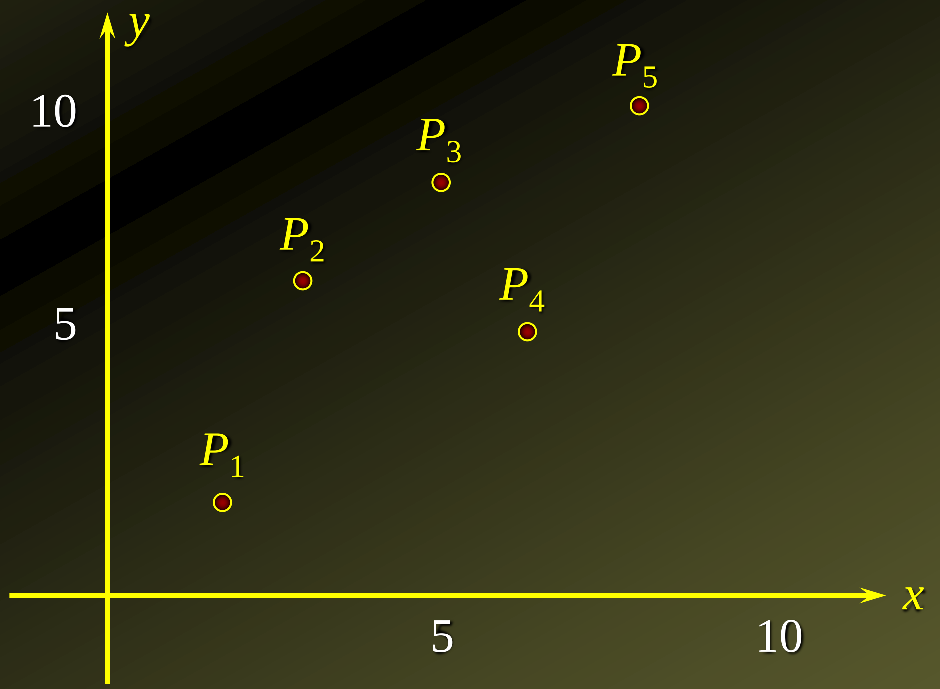


The Method of Least Squares

- Suppose we are given the data points

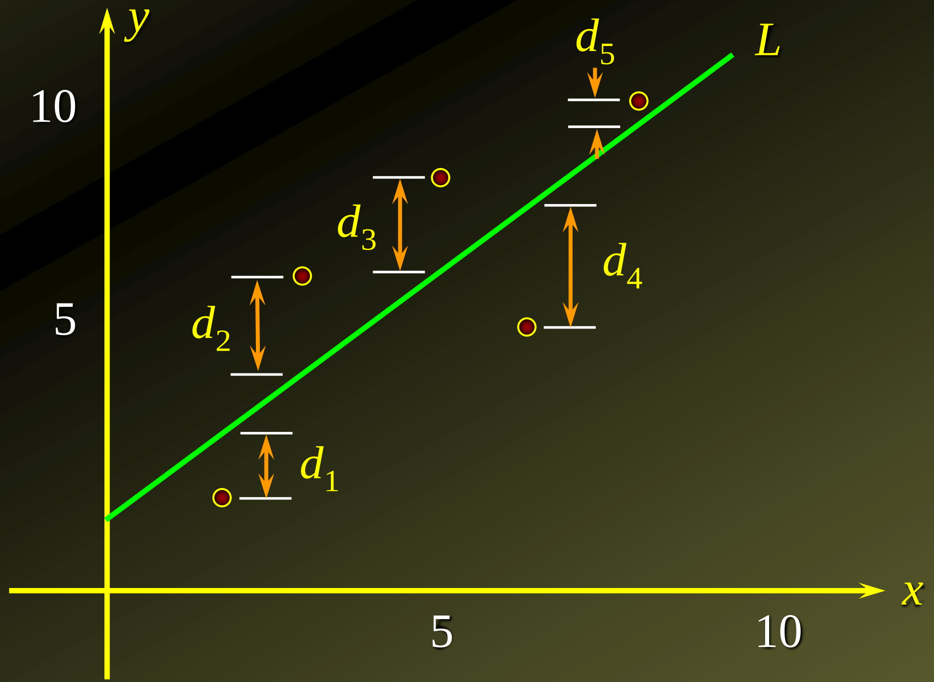
  $P_1(x_1, y_1)$, $P_2(x_2, y_2)$, $P_3(x_3, y_3)$, $P_4(x_4, y_4)$, and $P_5(x_5, y_5)$

  that describe the relationship between two variables x and y.
- By plotting these data points, we obtain a scatter diagram:



The Method of Least Squares

- Suppose we try to fit a straight line L to the data points P_1, P_2, P_3, P_4, and P_5.
- The line will miss these points by the amounts d_1, d_2, d_3, d_4, and d_5 respectively.

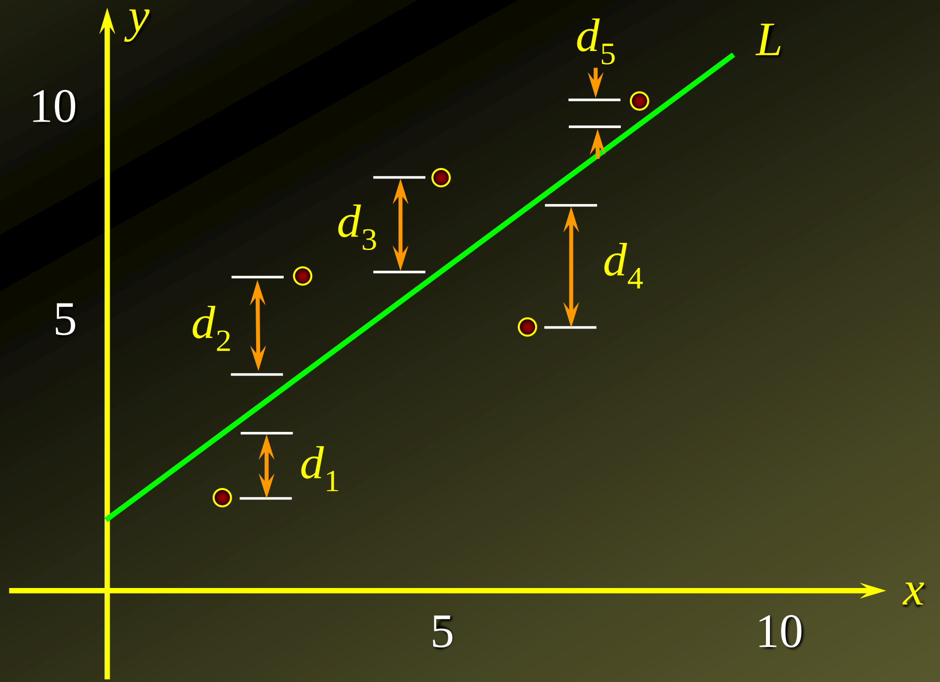


The Method of Least Squares

- The principle of least squares states that the straight line L that fits the data points best is the one chosen by requiring that the sum of the squares of d_1, d_2, d_3, d_4, and d_5, that is

$$d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2$$

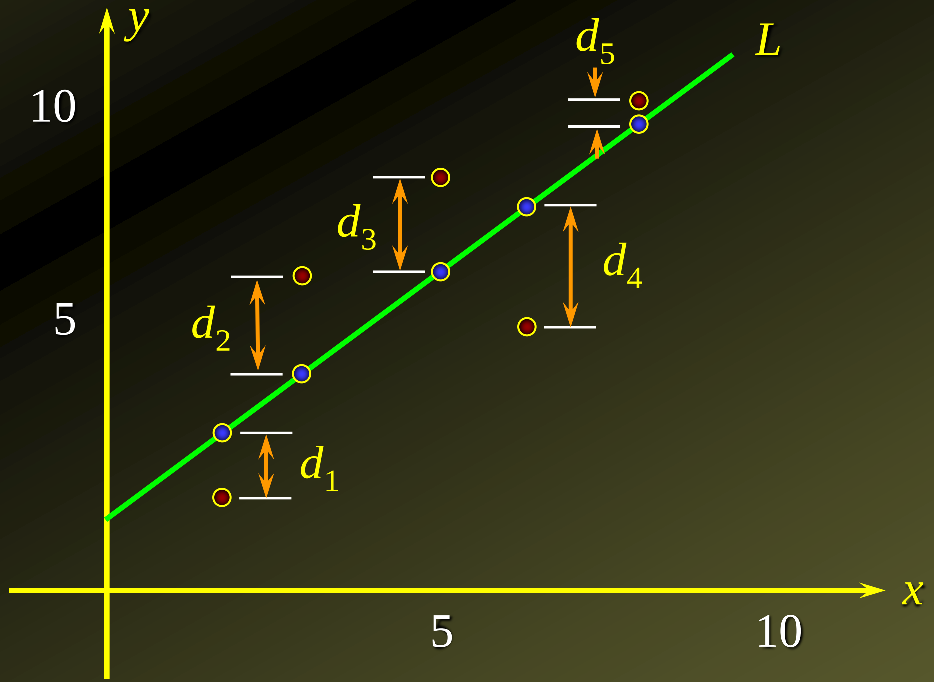
be made as small as possible.



The Method of Least Squares

- Suppose the **regression line** L is $y = f(x) = mx + b$, where m and b are **to be determined**.
- The distances d_1, d_2, d_3, d_4, and d_5, represent the **errors** the line L is making in estimating these points, so that

$d_1 = f(x_1) - y_1$, $d_2 = f(x_2) - y_2$, $d_3 = f(x_3) - y_3$, and so on.



The Method of Least Squares

- Observe that

$$
\begin{aligned}
d_1^2 + d_2^2 + d_3^2 + d_4^2 + d_5^2 \\
&= [f(x_1) - y_1]^2 + [f(x_2) - y_2]^2 + [f(x_3) - y_3]^2 \\
&\quad + [f(x_4) - y_4]^2 + [f(x_5) - y_5]^2 \\
&= [mx_1 + b - y_1]^2 + [mx_2 + b - y_2]^2 + [mx_3 + b - y_3]^2 \\
&\quad + [mx_4 + b - y_4]^2 + [mx_5 + b - y_5]^2
\end{aligned}
$$

- This may be viewed as a function of two variables m and b.
- Thus, the least-squares criterion is equivalent to minimizing the function

$$
\begin{aligned}
f(m,b) &= (mx_1 + b - y_1)^2 + (mx_2 + b - y_2)^2 + (mx_3 + b - y_3)^2 \\
&\quad + (mx_4 + b - y_4)^2 + (mx_5 + b - y_5)^2
\end{aligned}
$$

The Method of Least Squares

- We want to minimize

$$f(m,b) = (mx_1 + b - y_1)^2 + (mx_2 + b - y_2)^2 + (mx_3 + b - y_3)^2 + (mx_4 + b - y_4)^2 + (mx_5 + b - y_5)^2$$

- We first find the partial derivative with respect to *m*:

$$\frac{\partial f}{\partial m} = 2(mx_1 + b - y_1)x_1 + 2(mx_2 + b - y_2)x_2 + 2(mx_3 + b - y_3)x_3 + 2(mx_4 + b - y_4)x_4 + 2(mx_5 + b - y_5)x_5$$

$$= 2[mx_1^2 + bx_1 - y_1x_1 + mx_2^2 + bx_2 - y_2x_2 + mx_3^2 + bx_3 - y_3x_3 + mx_4^2 + bx_4 - y_4x_4 + mx_5^2 + bx_5 - y_5x_5]$$

$$= 2[(x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2)m + (x_1 + x_2 + x_3 + x_4 + x_5)b - (y_1x_1 + y_2x_2 + y_3x_3 + y_4x_4 + y_5x_5)]$$

The Method of Least Squares

- **We want to minimize**

$$f(m,b) = (mx_1 + b - y_1)^2 + (mx_2 + b - y_2)^2 + (mx_3 + b - y_3)^2 + (mx_4 + b - y_4)^2 + (mx_5 + b - y_5)^2$$

- **We now find the partial derivative with respect to b:**

$$\frac{\partial f}{\partial b} = 2(mx_1 + b - y_1) + 2(mx_2 + b - y_2) + 2(mx_3 + b - y_3) + 2(mx_4 + b - y_4) + 2(mx_5 + b - y_5)$$

$$= 2[(x_1 + x_2 + x_3 + x_4 + x_5)m + 5b - (y_1 + y_2 + y_3 + y_4 + y_5)]$$

The Method of Least Squares

- Setting

$$\frac{\partial f}{\partial m} = 0 \quad \text{and} \quad \frac{\partial f}{\partial b} = 0$$

gives

$$(x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2)m + (x_1 + x_2 + x_3 + x_4 + x_5)b$$
$$= y_1x_1 + y_2x_2 + y_3x_3 + y_4x_4 + y_5x_5$$

and

$$(x_1 + x_2 + x_3 + x_4 + x_5)m + 5b = y_1 + y_2 + y_3 + y_4 + y_5$$

- **Solving** the two **simultaneous equations** for m and b then leads to an equation $y = mx + b$.
- This equation will be the **'best fit' line**, or **regression line** for the given data points.

The Method of Least Squares

- Suppose we are given n **data points**:

$$P_1(x_1, y_1), P_2(x_2, y_2), P_3(x_3, y_3), \ldots , P_n(x_n, y_n)$$

- Then, the **least-squares (regression) line** for the data is given by the **linear equation**

$$y = f(x) = mx + b$$

where the **constants** m and b satisfy the equations

$$\boxed{\begin{aligned}(x_1^2 + x_2^2 + x_3^2 + \ldots + x_n^2)m + (x_1 + x_2 + x_3 + \ldots + x_n)b \\ = y_1x_1 + y_2x_2 + y_3x_3 + \ldots + y_nx_n\end{aligned}}$$

and

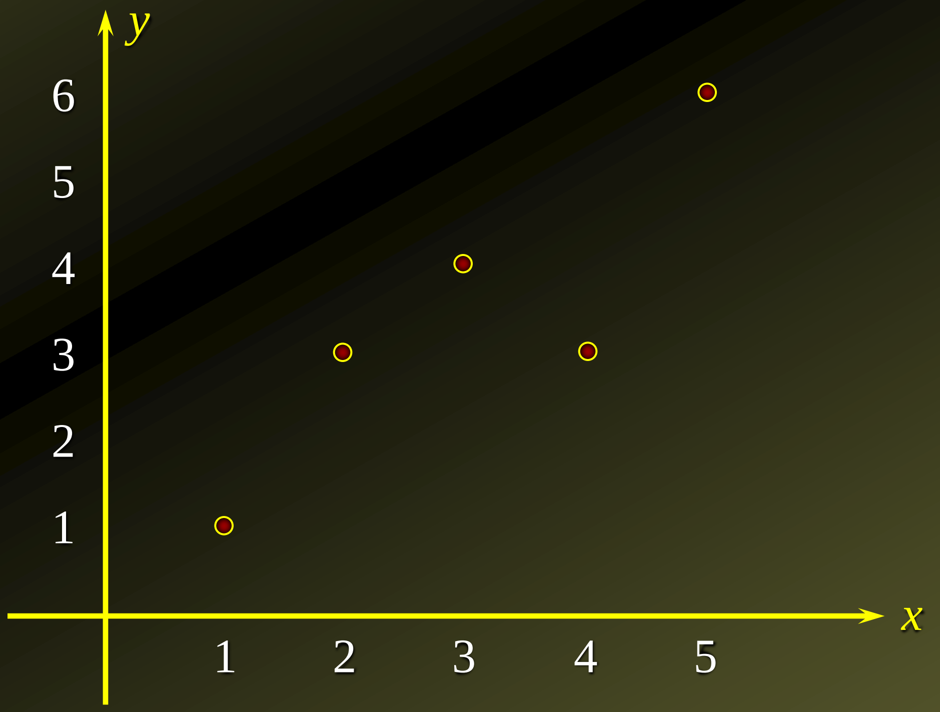
$$\boxed{(x_1 + x_2 + x_3 + \ldots + x_n)m + nb = y_1 + y_2 + y_3 + \ldots + y_n}$$

simultaneously.

- These last two equations are called **normal equations**.

Example

- Find the equation of the **least-squares line** for the data $P_1(1, 1)$, $P_2(2, 3)$, $P_3(3, 4)$, $P_4(4, 3)$, and $P_5(5, 6)$



Example

- **Find the equation of the least-squares line for the data**

  $P_1(1, 1)$, $P_2(2, 3)$, $P_3(3, 4)$, $P_4(4, 3)$, **and** $P_5(5, 6)$

Solution

- **Here, we have $n = 5$ and**

$x_1 = 1$	$x_2 = 2$	$x_3 = 3$	$x_4 = 4$	$x_5 = 5$
$y_1 = 1$	$y_2 = 3$	$y_3 = 4$	$y_4 = 3$	$y_5 = 6$

- **Substituting in the first equation we get**

$$(x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2)m + (x_1 + x_2 + x_3 + \dots + x_n)b$$
$$= y_1x_1 + y_2x_2 + y_3x_3 + \dots + y_nx_n$$

$$(1^2 + 2^2 + 3^2 + 4^2 + 5^2)m + (1 + 2 + 3 + 4 + 5)b$$
$$= (1)(1) + (3)(2) + (4)(3) + (3)(4) + (6)(5)$$

$$55m + 15b = 61$$

Example

- Find the equation of the least-squares line for the data

$$P_1(1, 1),\ P_2(2, 3),\ P_3(3, 4),\ P_4(4, 3),\ \text{and } P_5(5, 6)$$

Solution

- Here, we have $n = 5$ and

$$\begin{array}{lllll} x_1 = 1 & x_2 = 2 & x_3 = 3 & x_4 = 4 & x_5 = 5 \\ y_1 = 1 & y_2 = 3 & y_3 = 4 & y_4 = 3 & y_5 = 6 \end{array}$$

- Substituting in the second equation we get

$$(x_1 + x_2 + x_3 + \ldots + x_n)m + 5b = y_1 + y_2 + y_3 + \ldots + y_n$$

$$(1+2+3+4+5)m + 5b = 1+3+4+3+6$$

$$\underline{15m + 5b = 17}$$

Example

- Find the equation of the least-squares line for the data

$P_1(1, 1)$, $P_2(2, 3)$, $P_3(3, 4)$, $P_4(4, 3)$, and $P_5(5, 6)$

<u>**Solution**</u>

- Solving the simultaneous equations

$$55m + 15b = 61 \qquad 15m + 5b = 17$$

gives $m = 1$ and $b = 0.4$.

- Therefore, the required least-squares line is

$$y = x + 0.4$$

Example

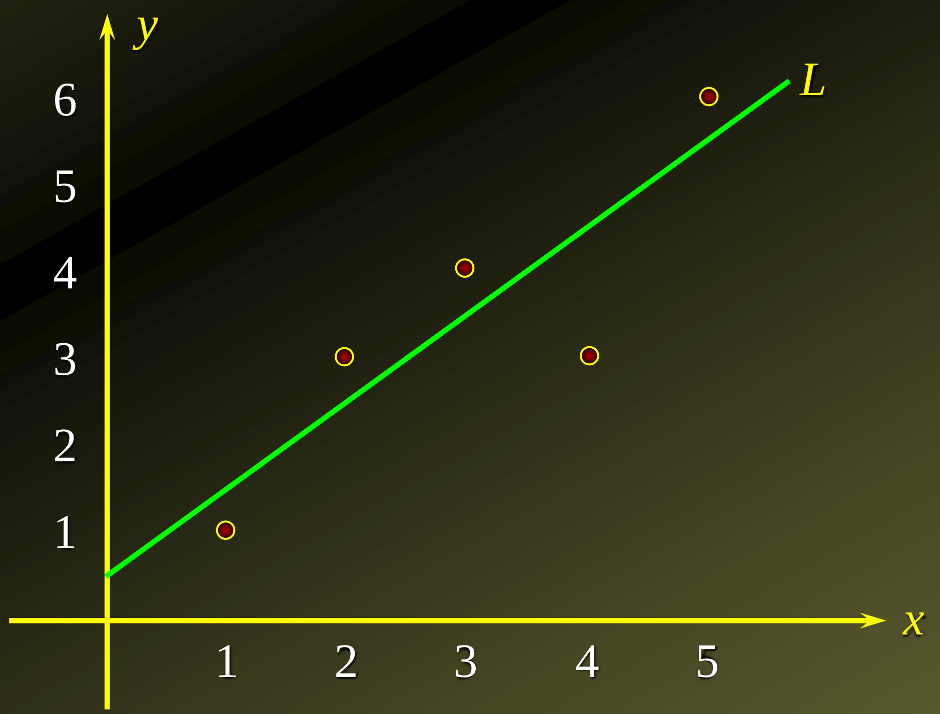
- Find the equation of the least-squares line for the data

$$P_1(1, 1),\ P_2(2, 3),\ P_3(3, 4),\ P_4(4, 3), \text{ and } P_5(5, 6)$$

Solution

- Below is the graph of the required least-squares line

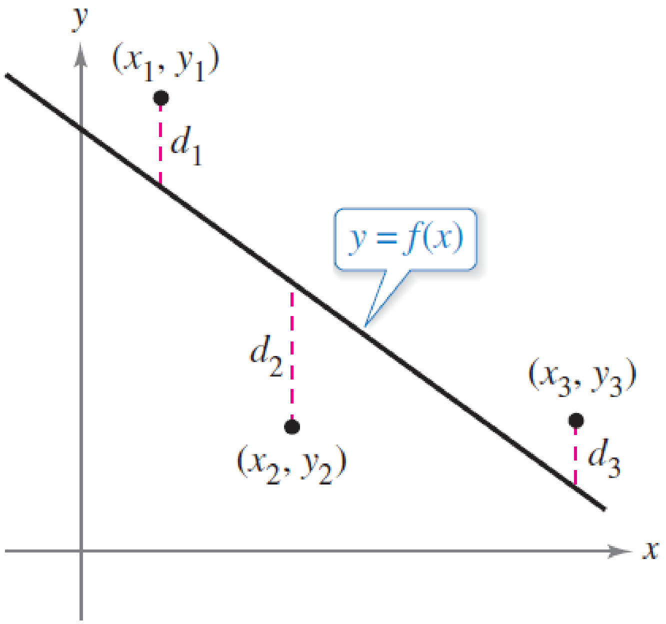
$$y = x + 0.4$$



The Method of Least Squares

- Graphically, S can be interpreted as the sum of the squares of the vertical distances between the graph of f and the given points in the plane, as shown in Figure 13.76.

- If the model is perfect, then $S = 0$. However, when perfection is not feasible, you can settle for a model that minimizes S.



Sum of the squared errors:
$S = d_1^2 + d_2^2 + d_3^2$

Figure 13.76

The Method of Least Squares

- For instance, the sum of the squared errors for the linear model in Figure 13.74 is $S = 17.6$.

- Statisticians call the *linear model* that minimizes S the **least squares regression line.**

- The proof that this line actually minimizes S involves the minimizing of a function of two variables.

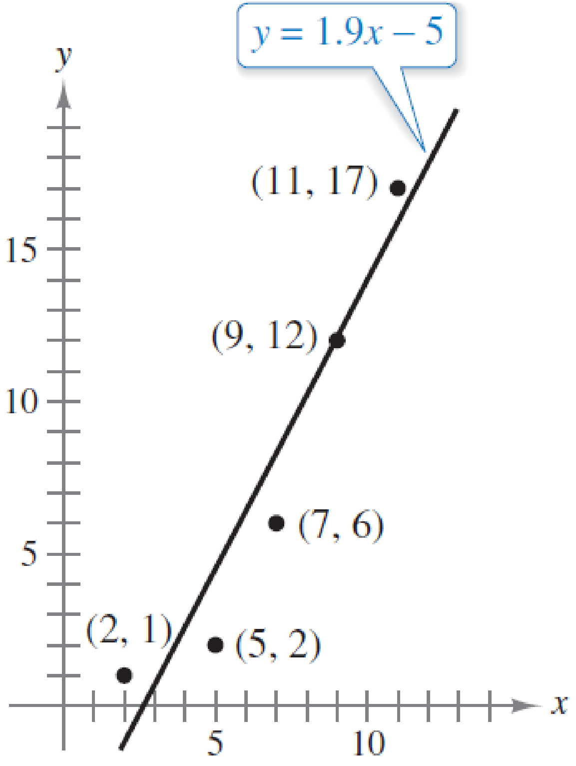


Figure 13.74

The Method of Least Squares

THEOREM 13.18 Least Squares Regression Line

The **least squares regression line** for $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ is given by $\boxed{f(x) = ax + b,}$ where

$$\boxed{a = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}} \quad \text{and} \quad \boxed{b = \frac{1}{n}\left(\sum_{i=1}^{n} y_i - a\sum_{i=1}^{n} x_i\right).}$$

The Method of Least Squares

- If the x-values are symmetrically spaced about the y-axis, then $\Sigma\, x_i = 0$ and the formulas for a and b simplify to

$$a = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} \qquad \text{and} \qquad b = \frac{1}{n}\sum_{i=1}^{n} y_i.$$

- This simplification is often possible with a translation of the x-values.

- For instance, given that the x-values in a data collection consist of the values 9, 10, 11, 12, and 13, you could let 11 be represented by 0.

Example 3 – *Finding the Least Squares Regression Line*

- Find the least squares regression line for the points (–3, 0), (–1, 1), (0, 2), and (2, 3).
- Solution:

✓ The table shows the calculations involved in finding the least squares regression line using n = 4.

x	y	xy	x^2
-3	0	0	9
-1	1	-1	1
0	2	0	0
2	3	6	4
$\sum_{i=1}^{n} x_i = -2$	$\sum_{i=1}^{n} y_i = 6$	$\sum_{i=1}^{n} x_i y_i = 5$	$\sum_{i=1}^{n} x_i^2 = 14$

Example 3 – *Solution*

cont'd

✓ Applying Theorem 13.18 produces

$$a = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} = \frac{4(5) - (-2)(6)}{4(14) - (-2)^2} = \frac{8}{13}$$

and

$$b = \frac{1}{n}\left(\sum_{i=1}^{n} y_i - a\sum_{i=1}^{n} x_i\right) = \frac{1}{4}\left[6 - \frac{8}{13}(-2)\right] = \frac{47}{26}.$$

Applied Example: Maximizing Profit

- A market research study provided the following data based on the projected monthly sales x (in thousands) of an adventure movie DVD.

p	38	36	34.5	30	28.5
x	2.2	5.4	7.0	11.5	14.6

- Find the demand equation if the demand curve is the least-squares line for these data.
- The total monthly cost function associated with producing and distributing the DVD is given by

$$C(x) = 4x + 25$$

where x denotes the number of discs (in thousands) produced and sold, and $C(x)$ is in thousands of dollars.
- Determine the unit wholesale price that will maximize monthly profits.

Applied Example: Maximizing Profit

Solution

- The calculations required for obtaining the normal equations may be summarized as follows:

x	p	x^2	xp
2.2	38.0	4.84	83.6
5.4	36.0	29.16	194.4
7.0	34.5	49.00	241.5
11.5	30.0	132.25	345.0
14.6	28.5	213.16	416.1
40.7	167.0	428.41	1280.6

- Thus, the nominal equations are

$$5b + 40.7m = 167 \quad \text{and} \quad 40.7b + 428.41m = 1280.6$$

Applied Example: Maximizing Profit

Solution

- **Solving the system of linear equations simultaneously, we find that**

$$m \approx -0.81 \quad \text{and} \quad b \approx 39.99$$

- **Therefore, the required demand equation is given by**

$$p = f(x) = -0.81x + 39.99 \qquad (0 \leq x \leq 49.37)$$

Applied Example: Maximizing Profit

Solution

- The **total revenue function** in this case is given by

$$\boxed{R(x) = xp}$$
$$= x(-0.81x + 39.99)$$
$$= -0.81x^2 + 39.99x$$

- Since the **total cost function** is

$$C(x) = 4x + 25$$

we see that the **profit function** is

$$\boxed{P(x) = R - C}$$
$$= -0.81x^2 + 39.99x - (4x + 25)$$
$$= -0.81x^2 + 35.99x - 25$$

Applied Example: Maximizing Profit

Solution

- To find the **absolute maximum** of $P(x)$ over the closed interval $[0, 49.37]$, we compute

$$P'(x) = -1.62x + 35.99$$

- Since $P(x) = 0$, we find that $x \approx 22.22$ as the **only critical point** of P.
- Finally, from the **table**

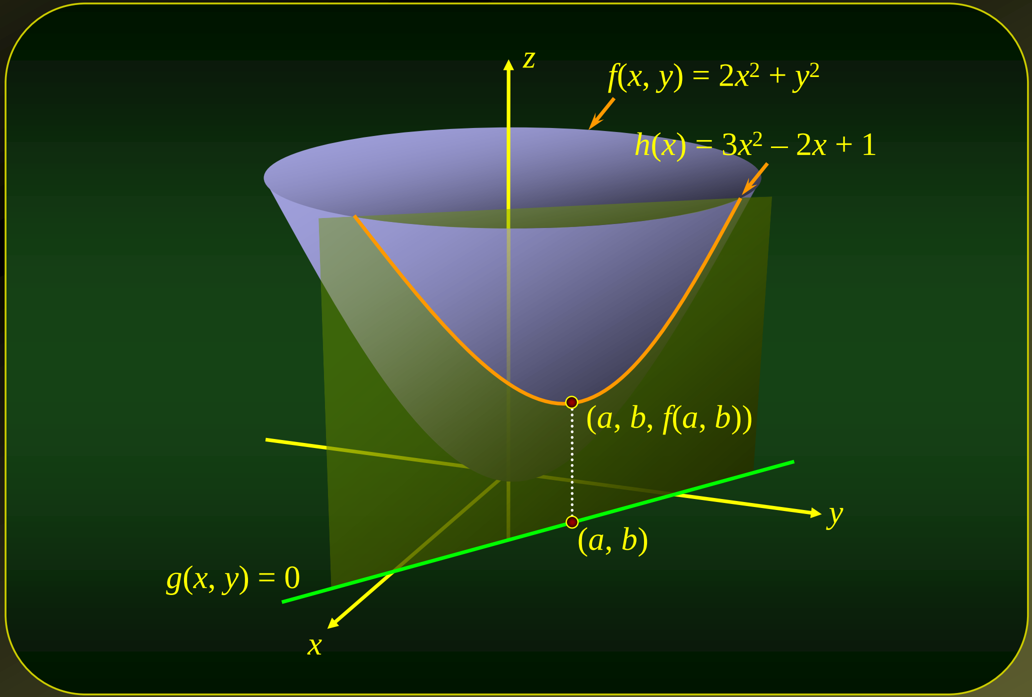
x	0	22.22	49.37
$P(x)$	–25	374.78	– 222.47

we see that the **optimal wholesale price** is

$$p = -0.81(22.22) + 39.99 = 21.99$$

or $21.99 per disc.

Constrained Maxima and Minima and the Method of Lagrange Multipliers



Constrained Maxima and Minima

- In many practical optimization problems, we must maximize or minimize a function in which the independent variables are subjected to certain further constraints.
- We shall discuss a powerful method for determining relative extrema of a function $f(x, y)$ whose independent variables x and y are required to satisfy one or more constraints of the form $\boxed{g(x, y) = 0.}$

Example

- Find the relative minimum of $f(x, y) = 2x^2 + y^2$ subject to the constraint $g(x, y) = x + y - 1 = 0$.

Solution

- Solving the constraint equation for y explicitly in terms of x, we obtain

$$y = -x + 1$$

- Substituting this value of y into $f(x, y)$ results in a function of x,

$$h(x) = 2x^2 + (-x+1)^2 = 3x^2 - 2x + 1$$

Example

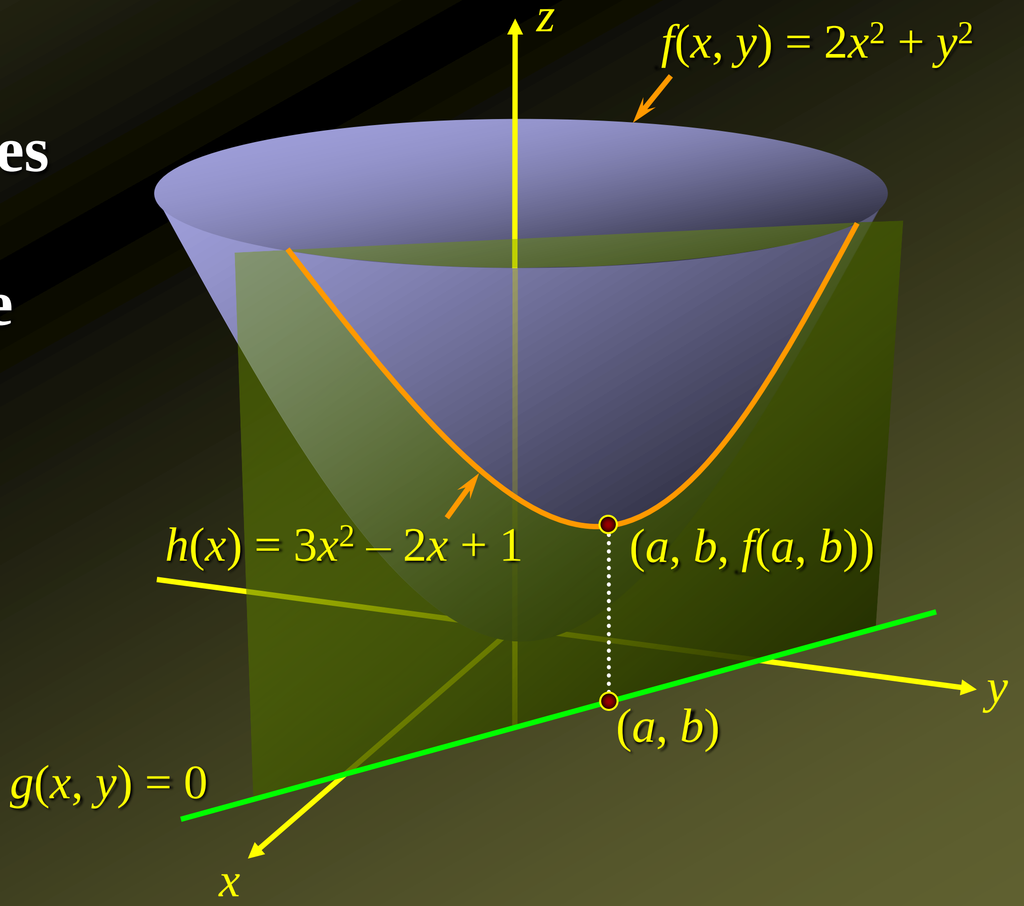
- Find the **relative minimum** of $f(x, y) = 2x^2 + y^2$ **subject to the constraint** $g(x, y) = x + y - 1 = 0$.

Solution

- **We have**

$$h(x) = 3x^2 - 2x + 1.$$

- **The function** h **describes the curve lying on the graph of** f **on which the constrained relative minimum point** (a, b) **of** f **occurs:**



Example

- Find the relative minimum of $f(x, y) = 2x^2 + y^2$ subject to the constraint $g(x, y) = x + y - 1 = 0$.

Solution

- To find this point (a, b), we determine the relative extrema of a function of one variable:

$$h'(x) = 6x - 2 = 2(3x - 1)$$

- Setting $h' = 0$ gives $x = \frac{1}{3}$ as the sole critical point of h.
- Next, we find $h''(x) = 6$ and, in particular, $h''(\frac{1}{3}) = 6 > 0$.
- Therefore, by the second derivative test, the point gives rise to a relative minimum of h.
- Substitute $x = \frac{1}{3}$ into the constraint equation $x + y - 1 = 0$ to get $y = \frac{1}{3}$.

Example

- Find the relative minimum of $f(x, y) = 2x^2 + y^2$ subject to the constraint $g(x, y) = x + y - 1 = 0$.

Solution

- Thus, the point $\left(\frac{1}{3}, \frac{2}{3}\right)$ gives rise to the required constrained relative minimum of f.

- Since $f\left(\frac{1}{3}, \frac{2}{3}\right) = 2\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{2}{3}$, the required constrained relative minimum value of f is $\frac{2}{3}$ at the point $\left(\frac{1}{3}, \frac{2}{3}\right)$.

- It may be shown that $\frac{2}{3}$ is in fact a constrained absolute minimum value of f.

The Method of Lagrange Multipliers

To find the relative extrema of the function $f(x, y)$ subject to the constraint $g(x, y) = 0$ (assuming that these extreme values exist),

1. Form an auxiliary function

$$\boxed{F(x, y, \lambda) = f(x, y) + \lambda g(x, y)}$$

   called the Lagrangian function (the variable λ is called the Lagrange multiplier).

2. Solve the system that consists of the equations

$$F_x = 0 \qquad F_y = 0 \qquad F_\lambda = 0$$

   for all values of x, y, and λ.

3. The solutions found in step 2 are candidates for the extrema of f.

Example

- Using the **method of Lagrange multipliers**, find the **relative minimum** of the function $f(x, y) = 2x^2 + y^2$ subject to the **constraint** $x + y = 1$.

Solution

- Write the **constraint equation**

$$x + y = 1$$

in the form

$$g(x, y) = x + y - 1 = 0$$

- Then, form the **Lagrangian function**

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

$$= (2x^2 + y^2) + \lambda(x + y - 1)$$

Example

- Using the method of Lagrange multipliers, find the relative minimum of the function $f(x, y) = 2x^2 + y^2$ subject to the constraint $x + y = 1$.

Solution

- We have $F = 2x^2 + y^2 + \lambda(x + y - 1)$
- To find the critical point(s) of the function F, solve the system composed of the equations

$$F_x = 4x + \lambda = 0 \qquad F_y = 2y + \lambda = 0 \qquad F_\lambda = x + y - 1 = 0$$

- Solving the first and second equations for x and y in terms of λ, we obtain

$$x = -\frac{1}{4}\lambda \quad \text{and} \quad y = -\frac{1}{2}\lambda$$

- Which, upon substitution into the third equation yields

$$-\frac{1}{4}\lambda - \frac{1}{2}\lambda - 1 = 0 \quad \text{or} \quad \lambda = -\frac{4}{3}$$

Example

- Using the **method of Lagrange multipliers**, find the **relative minimum** of the function $f(x, y) = 2x^2 + y^2$ subject to the **constraint** $x + y = 1$.

Solution

- Substituting $\lambda = -\frac{4}{3}$ into $x = -\frac{1}{4}\lambda$ and $y = -\frac{1}{2}\lambda$

  yields $x = -\frac{1}{4}(-\frac{4}{3}) = \frac{1}{3}$ and $y = -\frac{1}{2}(-\frac{4}{3}) = \frac{2}{3}$

- Therefore, $x = \frac{1}{3}$ and $y = \frac{2}{3}$, and $\left(\frac{1}{3}, \frac{2}{3}\right)$ results in a **constrained minimum** of the function f.

Applied Example: Designing a Cruise-Ship Pool

- The operators of the Viking Princess, a luxury cruise liner, are contemplating the addition of another swimming pool to the ship.
- The chief engineer has suggested that an area of the form of an ellipse located in the rear of the promenade deck would be suitable for this purpose.
- It has been determined that the shape of the ellipse may be described by the equation

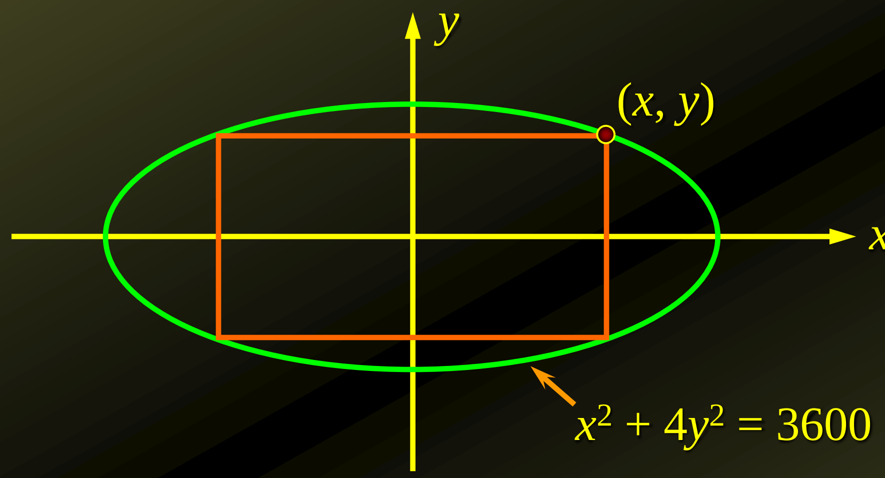
  $$x^2 + 4y^2 = 3600$$

  where x and y are measured in feet.
- Viking's operators would like to know the dimensions of the rectangular pool with the largest possible area that would meet these requirements.

Applied Example: Designing a Cruise-Ship Pool

Solution

- We want to maximize the area of the rectangle that will fit the ellipse:



- Letting the sides of the rectangle be $2x$ and $2y$ feet, we see that the area of the rectangle is $A = 4xy$.
- Furthermore, the point (x, y) must be constrained to lie on the ellipse so that it satisfies the equation $x^2 + 4y^2 = 3600$.

Applied Example: Designing a Cruise-Ship Pool

Solution

- **Thus, the problem is equivalent to the problem of maximizing the function**

$$\boxed{f(x, y) = 4xy}$$

subject to the constraint

$$g(x, y) = x^2 + 4y^2 - 3600 = 0$$

- **The Lagrangian function is**

$$F(x, y, \lambda) = f(x, y) + \lambda g(x, y) = 4xy + \lambda(x^2 + 4y^2 - 3600)$$

- **To find the critical points of F, we solve the system of equations**

$$F_x = 4y + 2\lambda x = 0$$

$$F_y = 4x + 8\lambda y = 0$$

$$F_\lambda = x^2 + 4y^2 - 3600 = 0$$

Applied Example: Designing a Cruise-Ship Pool

Solution

- We have
$$F_x = 4y + 2\lambda x = 0$$
$$F_y = 4x + 8\lambda y = 0$$
$$F_\lambda = x^2 + 4y^2 - 3600 = 0$$

- Solving the first equation for λ, we obtain $\lambda = -\dfrac{2y}{x}$

- Which, substituting into the second equation, yields

$$4x + 8\left(-\frac{2y}{x}\right)y = 0 \qquad \text{or} \qquad x^2 - 4y^2 = 0$$

- Solving for x yields $x = \pm 2y$.

Applied Example: Designing a Cruise-Ship Pool

Solution

- We have

$$F_x = 4y + 2\lambda x = 0$$
$$F_y = 4x + 8\lambda y = 0$$
$$F_\lambda = x^2 + 4y^2 - 3600 = 0$$

- Substituting $x = \pm 2y$ into the third equation, we have

$$4y^2 + 4y^2 - 3600 = 0$$

- Which, upon solving for y yields

$$y = \pm\sqrt{450} = \pm 15\sqrt{2}$$

- The corresponding values of x are

$$x = \pm 2y = \pm 2(15\sqrt{2}) = \pm 30\sqrt{2}$$

- Since both x and y must be nonnegative, we have

$$x = 30\sqrt{2} \quad \text{and} \quad y = 15\sqrt{2}$$

or approximately 42 X 85 feet.